

9/24/2025	11.30-13.30 B108	1	2	AL	La cellula e le principali biomolecole
9/24/2025	14.30-16.30 A203	2	2	AL	Il genoma: struttura e funzione
9/29/2025	11:30 - 13:30 A203	3	2	AL	Trascrizione: RNA codificante e non-codificante
10/1/2025	11.30-13.30 B108	4	2	AL	Traduzione sintesi e struttura delle proteine
10/1/2025	14.30-16.30 A203	5	2	AL	Le proteine: struttura, segnalazione e metabolismo cellulare
10/6/2025	11:30 - 13:30 A203	6	2	AL	Motilità intra ed extra cellulare
10/8/2025	11.30-13.30 B108	7	2	AL	Interazioni cellulari
10/8/2025	14.30-16.30 A203	8	2	AL	Summary - Concepts revisions - Questions
10/13/2025	11:30 - 13:30 A203	9	2	AL	
10/15/2025	11.30-13.30 B108	10	2	AL	Divisione cellulare: replicazione del genoma cellulare, mitosi e meiosi
					Fedeltà della informazione genetica: meccanismi di controllo della stabilità del
10/15/2025	14.30-16.30 A203	11	2	AL	genoma, instabilità genomica e patologie genetiche
10/20/2025	11:30 - 13:30 A203	12	2	AL	Biotecnologie: Ingegneria Genetica e Gene Editing
10/22/2025	11.30-13.30 B108	13	2	AL	Le centrali energetiche cellulari: I mitocondri e la catena respiratoria
10/22/2025	14.30-16.30 A203	14	2	AL	Morte cellulare: apoptosi, ferroptosi, autolisi e necrosi
10/29/2025	11.30-13.30 B108	15	2	AL	Summary - Concepts revisions - Questions
10/29/2025	14.30-16.30 A203	16	2	SB	Osso-tendini -muscoli-NMJ
11/5/2025	11.30-13.30 B108	17	2	AL	First written test
11/5/2025	14.30-16.30 A203	18	2	SB	Principles of neurogenesis
11/12/2025	11.30-13.30 B108	19	2	SB	Sistema nervoso periferico/autonomico
11/12/2025	14.30-16.30 A203	20	2	SB	Sistemi sensoriali
11/19/2025	11.30-13.30 B108	21	2	SB	Visione
11/19/2025	14.30-16.30 A203	22	2	SB	Sistema cardiovascolare
11/26/2025	11.30-13.30 B108	23	2	SB	Sistema gastro-intestinale
11/26/2025	14.30-16.30 A203	24	2	SB	Sistema endocrino/esocrino
12/3/2025	11.30-13.30 B108	25	2	SB	Sistema immunitario 1
12/3/2025	14.30-16.30 A203	26	2	SB	Sistema immunitario 2
12/10/2025	11.30-13.30 B108		2	SB	Backup
12/10/2025	14.30-16.30 A203	27	2	AL	Laboratorio di istologia patologica
12/17/2025	11.30-13.30 B108	28	2	SB	Second written test

The cell: basic unit of life

Cell by definition

Cell, in biology, is the basic membrane-bound unit that contains the fundamental molecules of life and of which all living things are composed.



The cell is the basic structural and functional unit of all forms of life. Every cell consists of cytoplasm enclosed within a membrane.

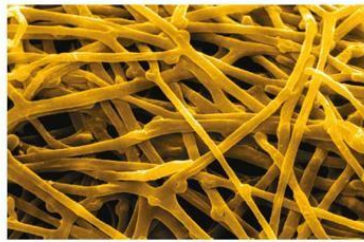


The smallest unit that can live on its own and that makes up all living organisms. A cell has three main parts: the cell membrane, *the genetic material (instead of nucleus)*, and the cytoplasm.



The cell: basic unit of life

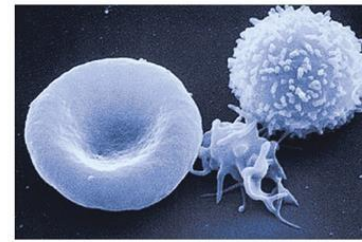
La cellula e' l'unita' di base della struttura di tutti gli organismi
Tutti gli organismi consistono di una o piu' cellule



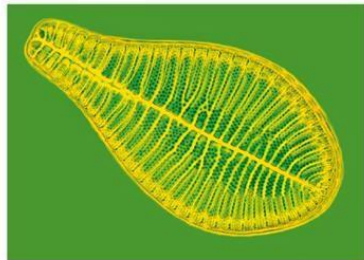
(a) Cellule filamentoze dei funghi 5 μm



(b) Cellula batterica di *Treponema* 5 μm



(c) Globuli rossi umani e una piastrina 5 μm



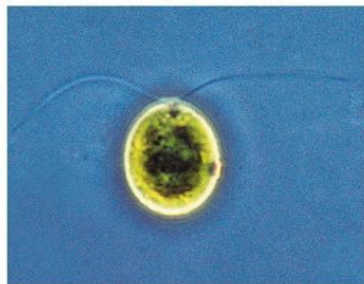
(d) *Podocystis* (una diatomea) 50 μm



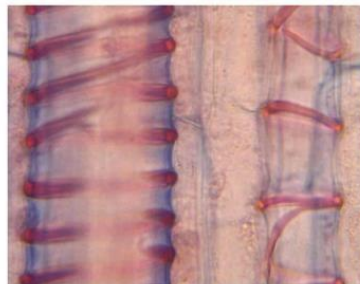
(e) *Stentor* (un protozoo) 100 μm



(f) Oocita e spermatozoi umani 50 μm



(g) *Chlamydomonas* (un'alga) 10 μm



(h) Cellule di xilema di pianta 100 μm



(i) Un neurone della retina 50 μm

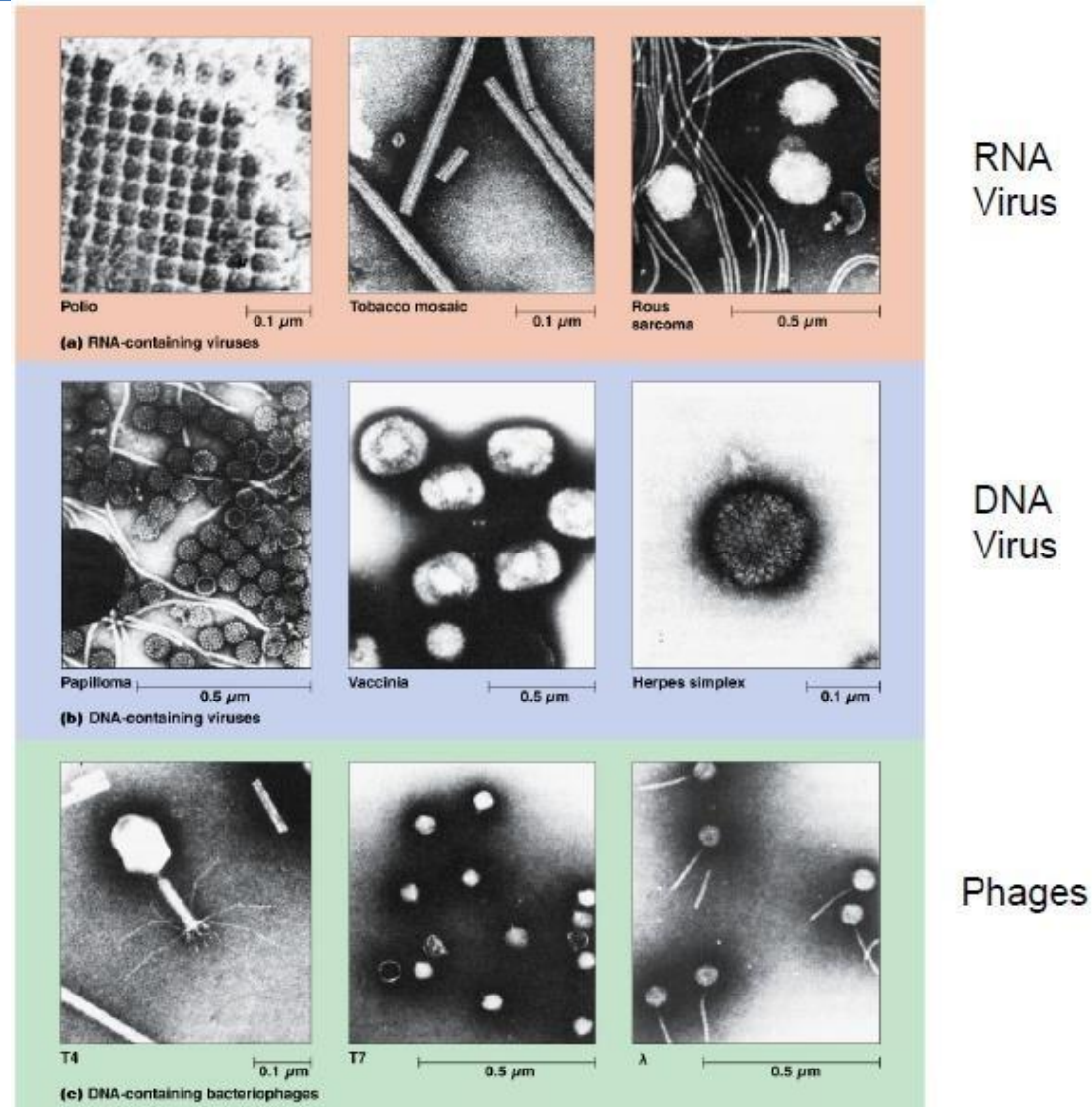
The cell: basic unit of life

Virus ARE NOT cells!!

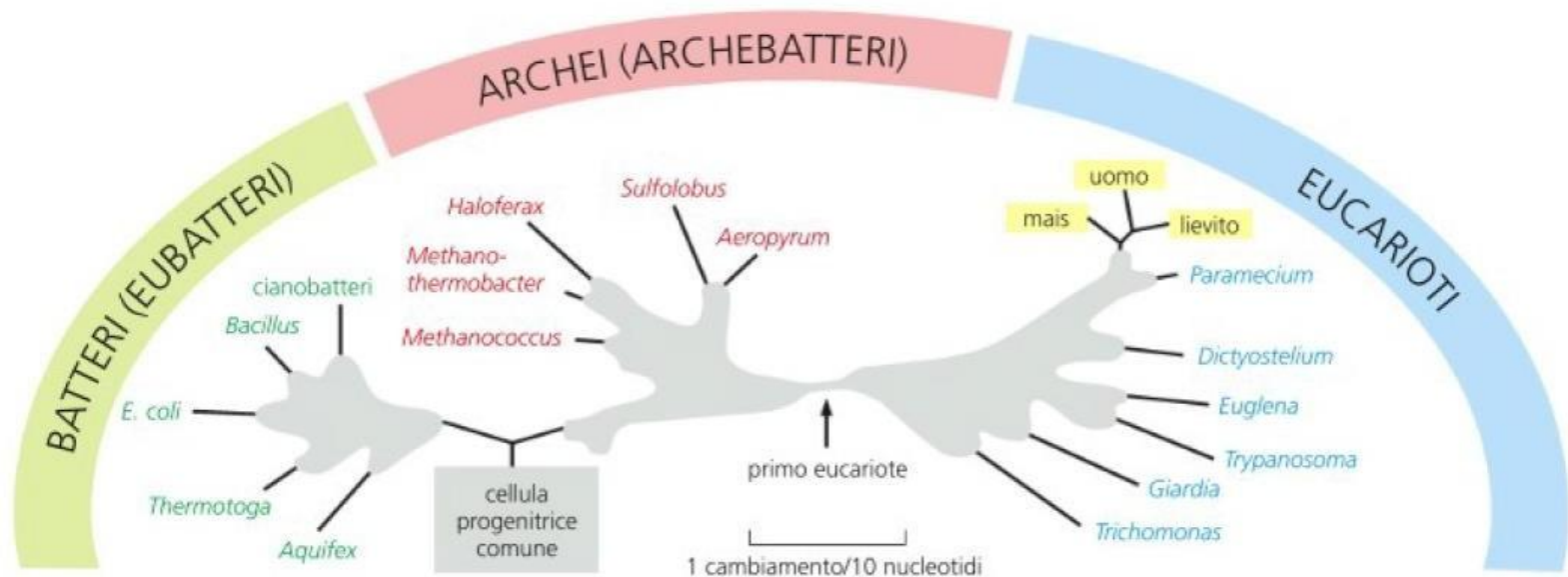
A virus is a **submicroscopic infectious agent that replicates only inside the living cells of an organism.**[1]

Viruses infect all types of life forms, from animals and plants to microorganisms, including bacteria and archaea.[2]

From Wikipedia



The cell: basic unit of life



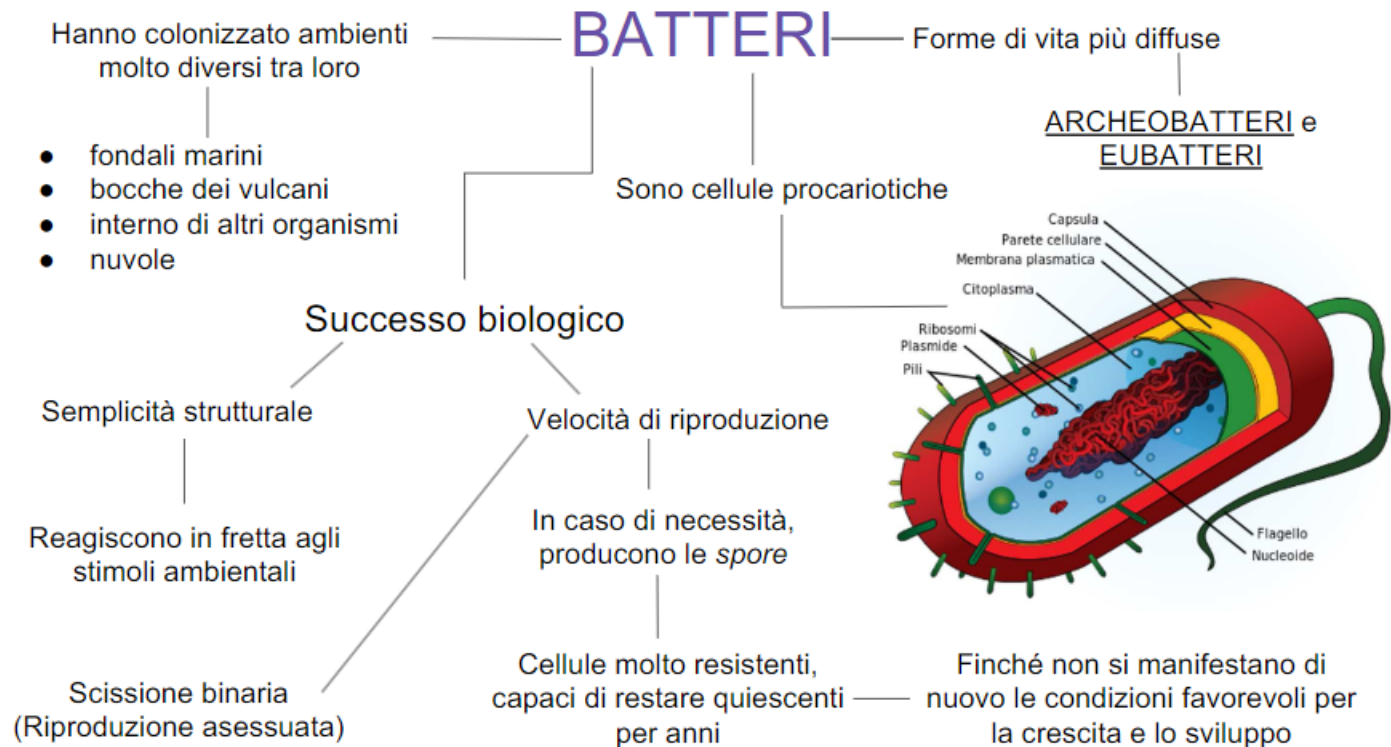
Cells can be classified as Prokaryotes or Eukaryotes

- **Prokaryotes:** the simpler type of cell, typical of bacteria (Eubacteria and Archaeobacteria)
- **Eukaryotes:** the complex type of cell, typical of algae, protozoa, fungi, plants, and animals.

The cell: basic unit of life

Prokaryotes

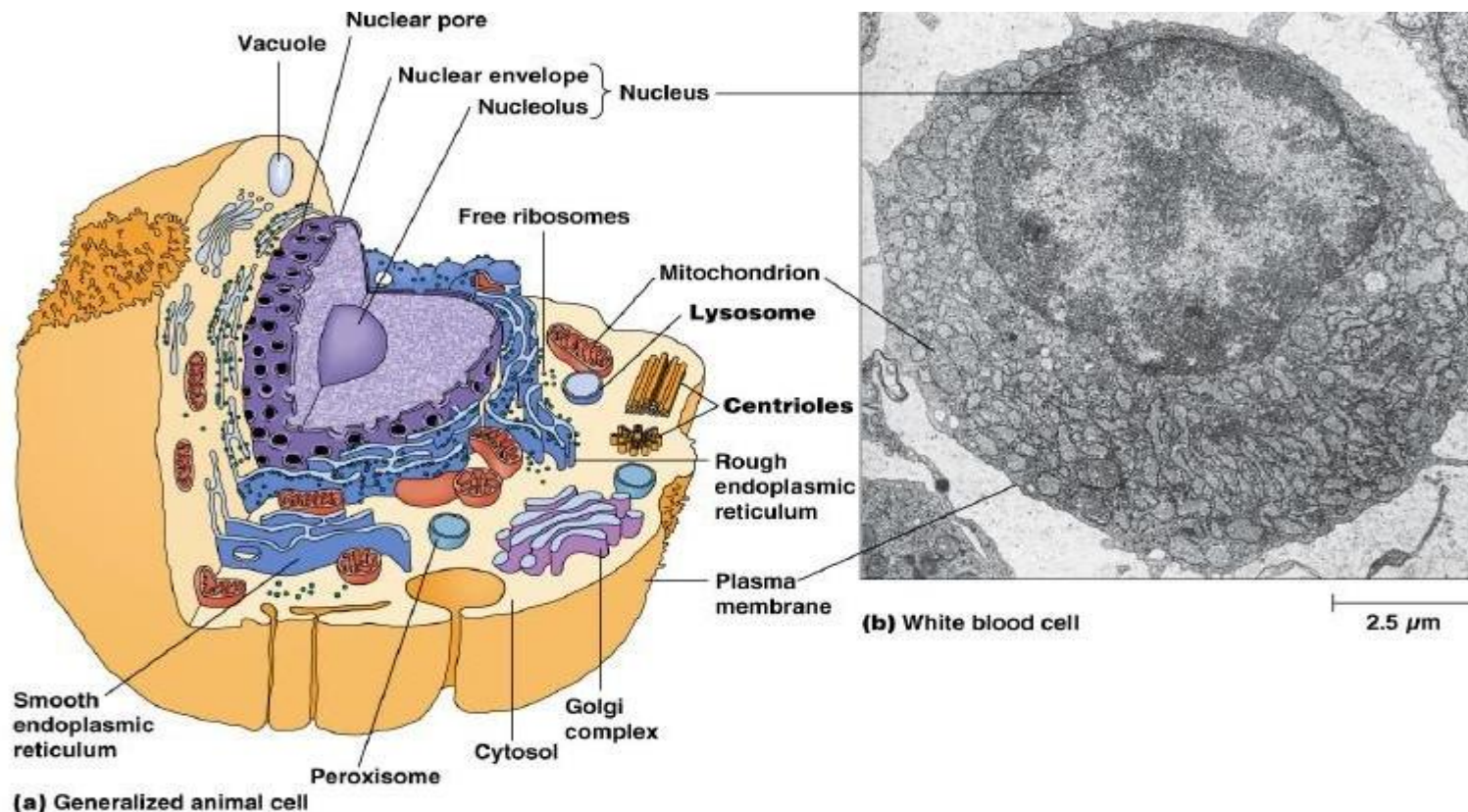
Le prime entità definibili come cellule si calcola siano comparse circa 3,5 miliardi di anni fa. I procarioti (Eubatteri e Archeobatteri) presentano struttura cellulare primitiva composta da capsula saccaridica, parete di peptidoglicani, membrana cellulare, citoplasma, materiale genetico libero o nucleotide.



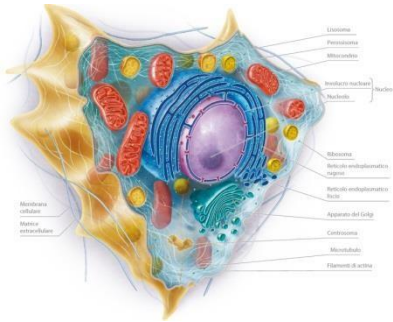
The cell: basic unit of life

Eukaryotes

Eukaryotes make extensive use of internal membranes to compartmentalize specific functions inside organelles: nucleus (*eu*=vero e *karyon*=nucleo), endoplasmic reticulum, Golgi complex, mitochondria, chloroplasts, lysosomes, peroxisomes and various types of vacuoles and vesicles.



The cell: basic unit of life

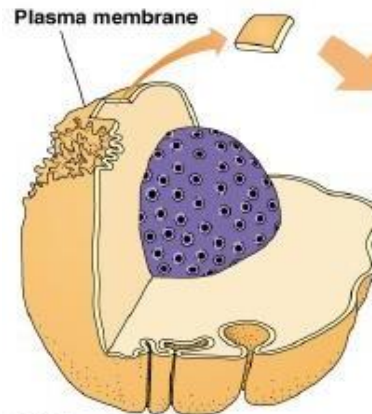


Cell Membrane

Proteins with a variety of roles:

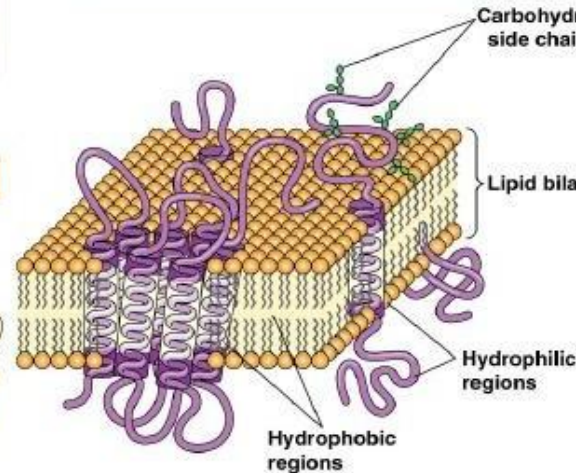
- **Enzymes:** catalyze reactions associated with the membranes
- **Anchors** for structural components of the cytoskeleton
- **Transport proteins** move substances across the membrane
- **Receptors** for external signals trigger processes within the cell
- The latter two are **transmembrane proteins**

(a) A cell. A cutaway view of an animal cell, showing the orientation of the piece of membrane shown in part b.

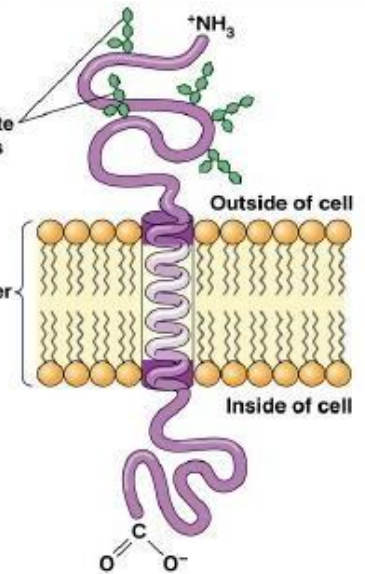


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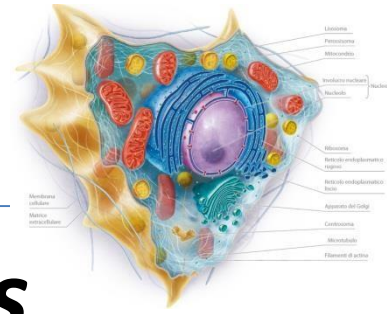
(b) Plasma membrane with membrane proteins. The plasma membrane consists of a lipid bilayer with membrane proteins suspended in it. Their hydrophobic regions are associated with the interior of the bilayer and their hydrophilic regions protrude from the membrane on one or both sides of the bilayer.



(c) Lipid bilayer with a glycoprotein. Most membrane proteins have at least one hydrophobic membrane-spanning domain. Proteins in the plasma membrane are typically glycoproteins with short carbohydrate side chains attached to the protein on the external side of the membrane.

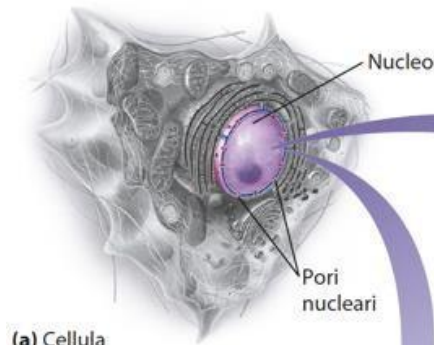


The cell: basic unit of life

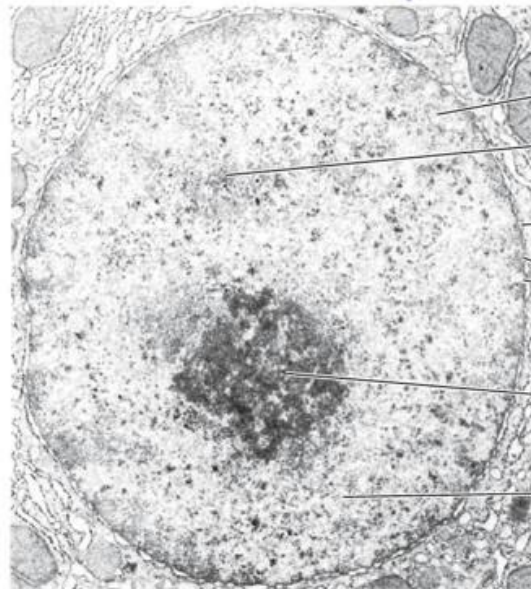


The Nucleus

- It *contains the DNA* and is surrounded by the *nuclear envelope*, composed of inner and outer membranes
- The *nuclear envelope* has *numerous openings called pores*, each of which is a transport channel, lined with a nuclear pore complex

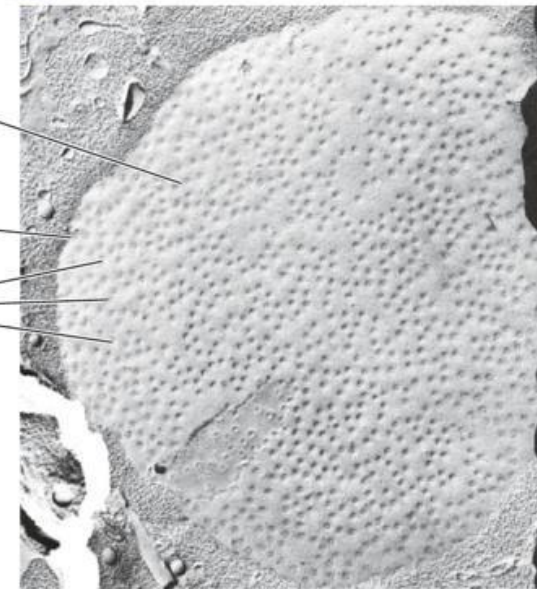


(a) Cellula



(b) Nucleo

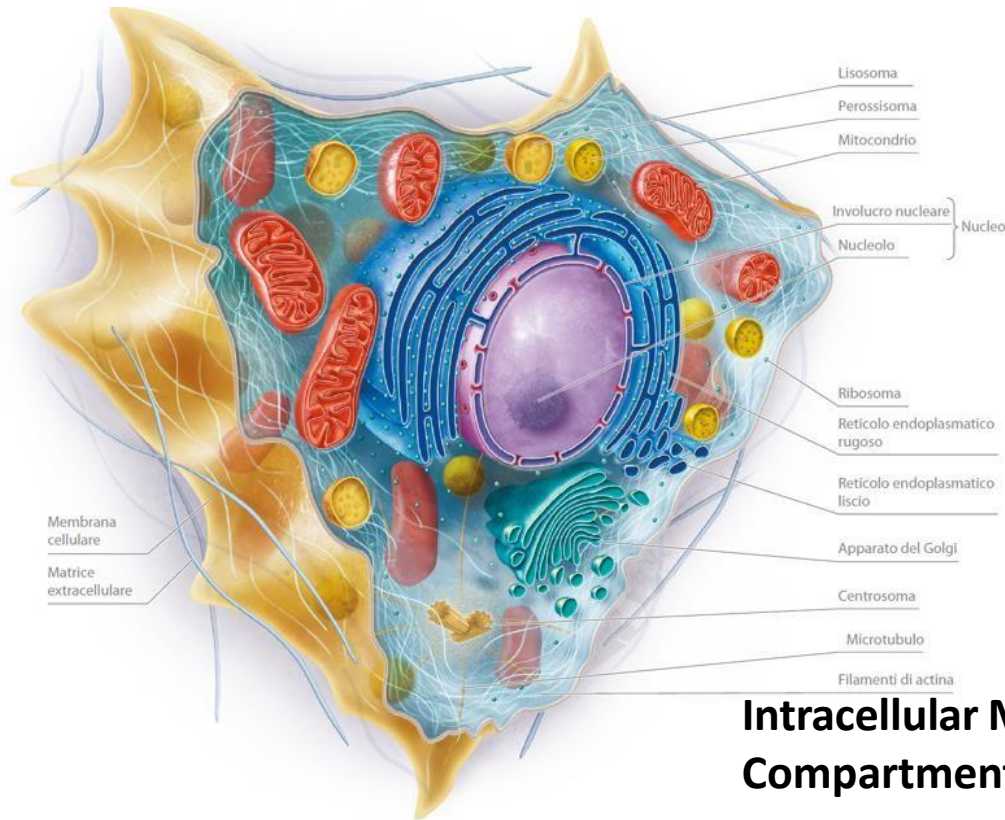
1 μm



(c) Involucro nucleare

2,5 μm

The cell: basic unit of life



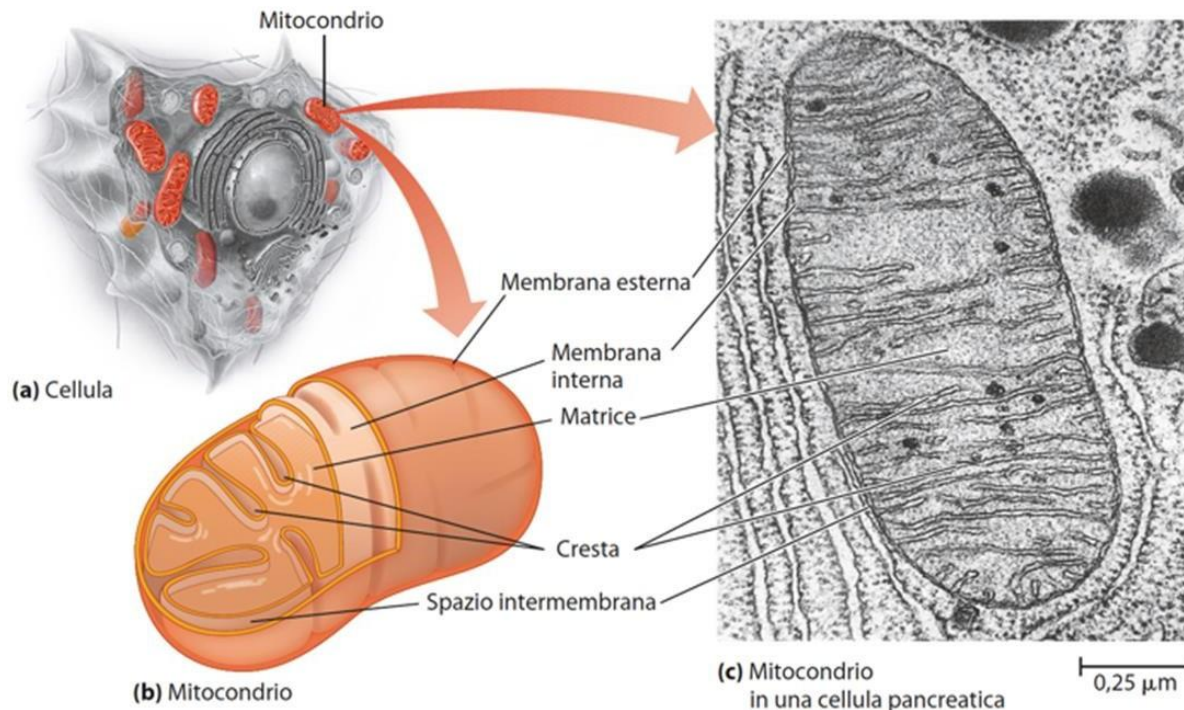
Organelles

Intracellular Membranes and Organelles Define Compartments

- The internal volume of the cell outside the nucleus is called the *cytoplasm* and is occupied by *organelles* and the semifluid *cytosol*
- A number of heritable human disorders are caused by malfunctions of specific organelles

The cell: basic unit of life

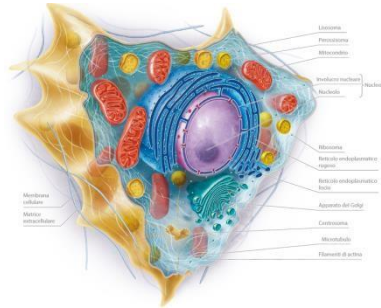
The mitochondrion



Mitochondria, found in all eukaryotic cells, are the site of aerobic respiration

- They are comparable in size to bacteria
- Most eukaryotic cells contain *hundreds of mitochondria*, each of which is surrounded by an inner and outer mitochondrial membrane

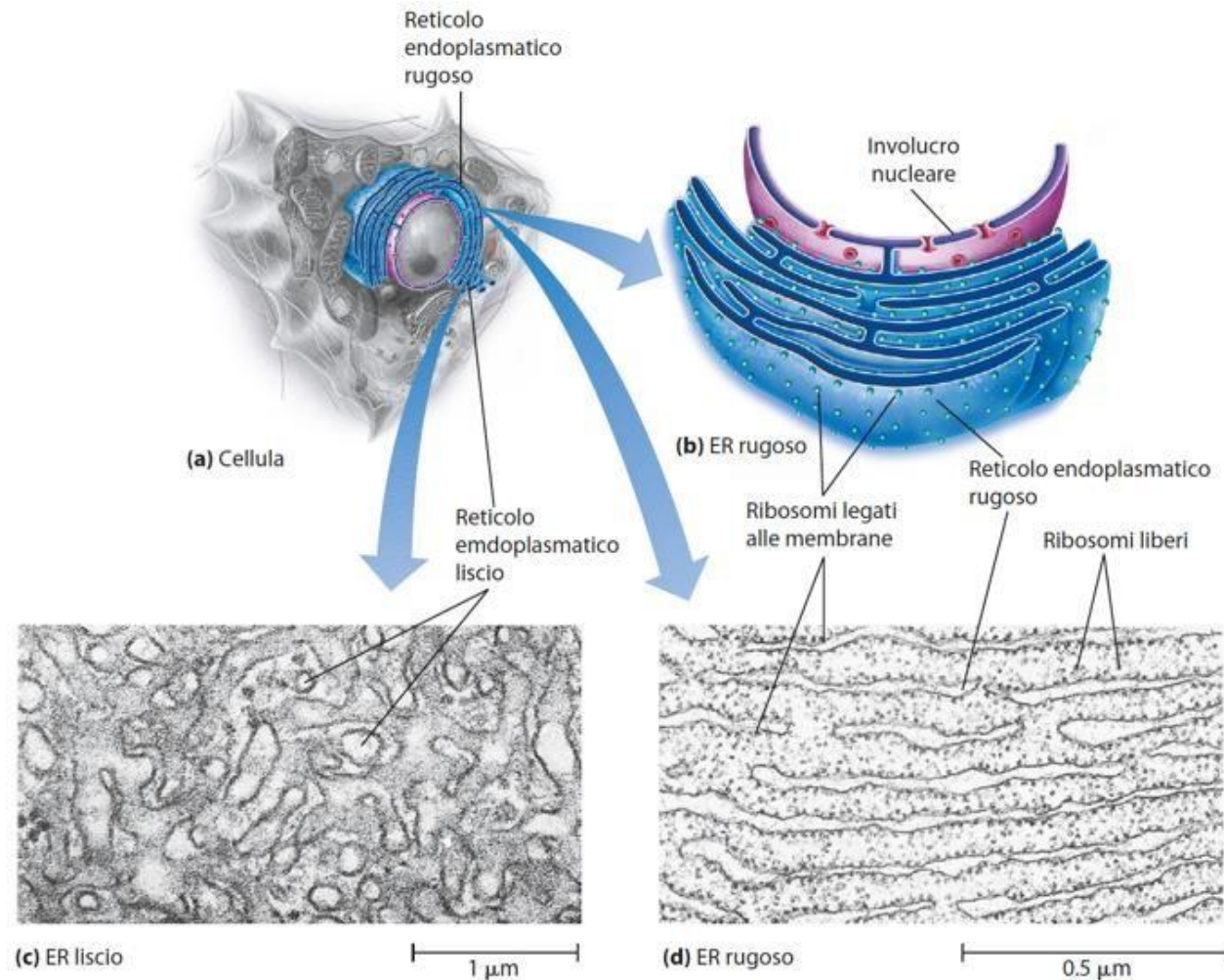
The cell: basic unit of life



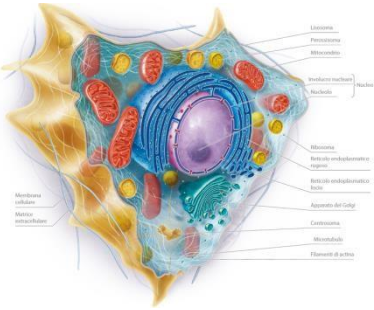
Endoplasmic reticulum

Almost every eukaryotic cell has a network of membranes in the cytoplasm, called the **endoplasmic reticulum (ER)**

- It consists of tubular membranes and flattened sacs called **cisternae**
- The internal space of the ER is called the **lumen**
- The ER is continuous with the other membranes in the cell

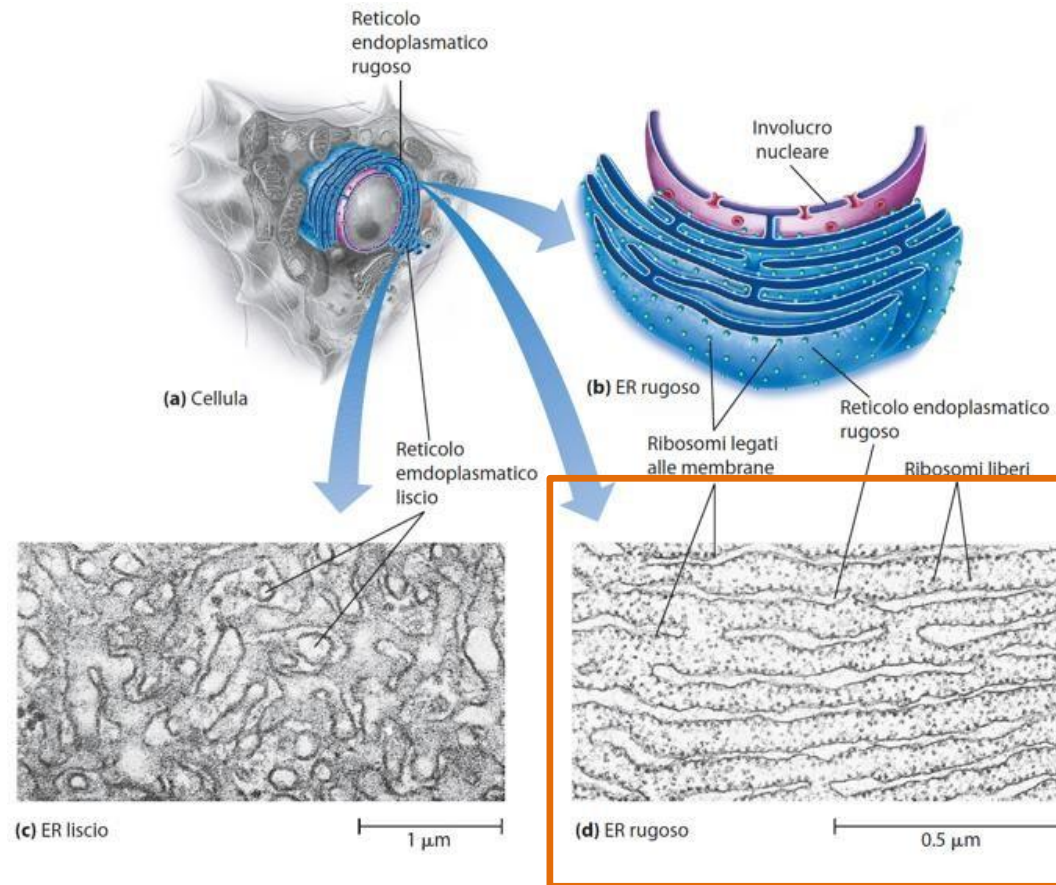


The cell: basic unit of life

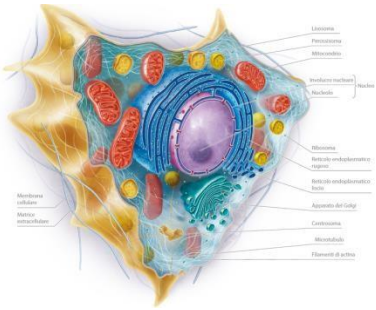


Endoplasmic reticulum

- ER can be *rough* or *smooth* in appearance
- *Rough ER is studded with ribosomes* on the cytoplasmic side of the membrane
- These *ribosomes synthesize polypeptides that accumulate within* the membrane or are transported across it to the lumen
- Free ribosomes are not associated with the ER

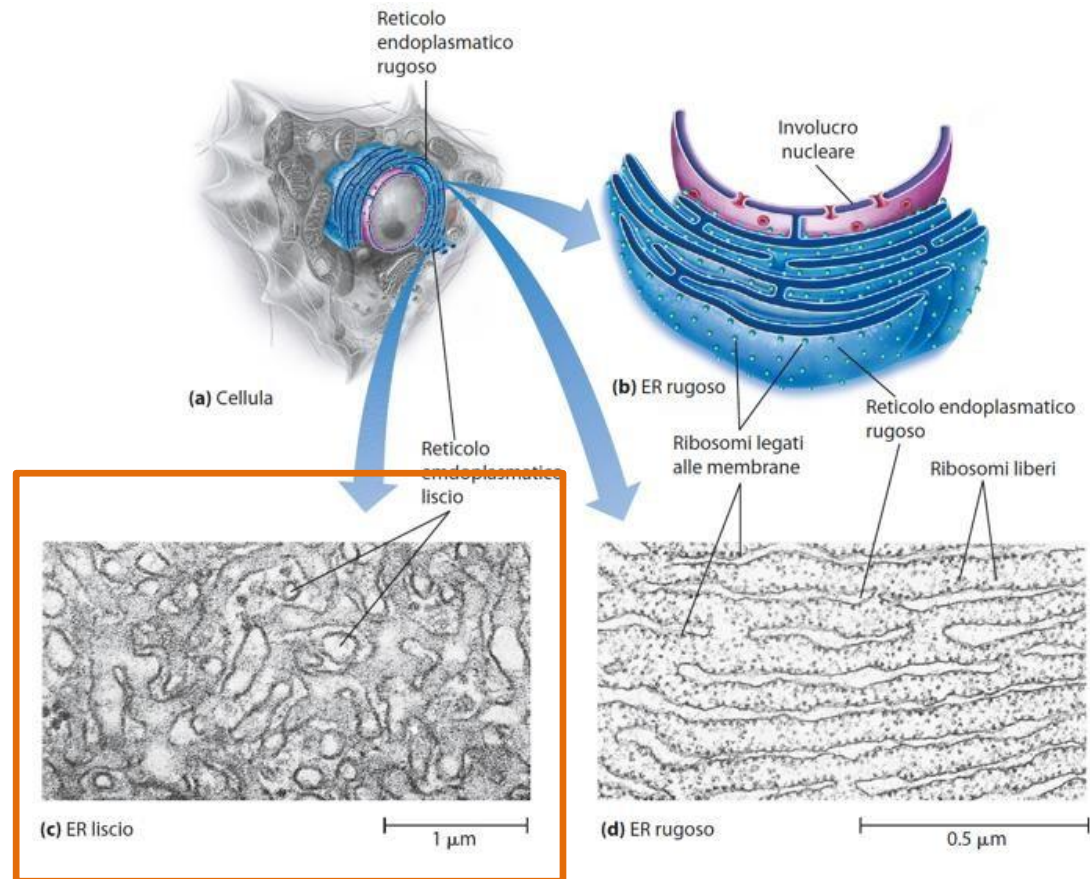


The cell: basic unit of life



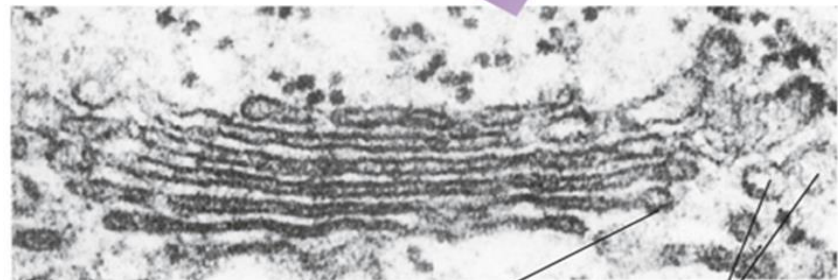
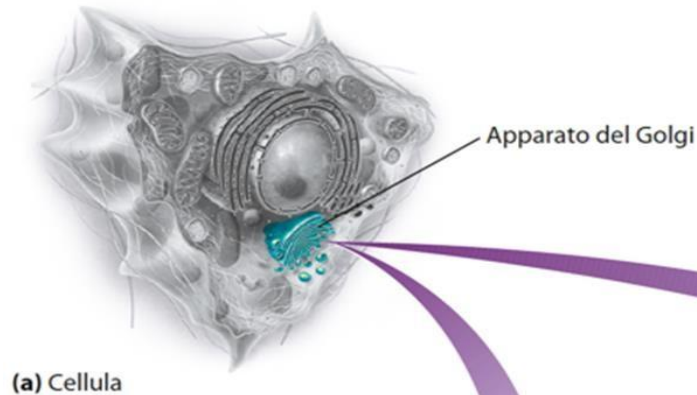
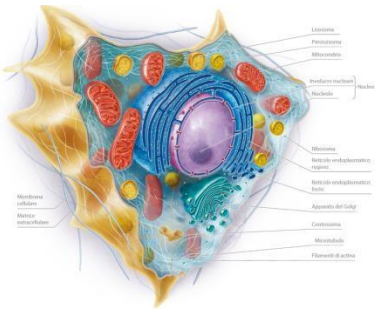
Endoplasmic reticulum

- **Smooth ER** has no role in protein synthesis
- It is involved in the synthesis of lipids and steroids such as cholesterol and its derivatives
- Smooth ER is responsible for inactivating and detoxifying potentially harmful substances



The cell: basic unit of life

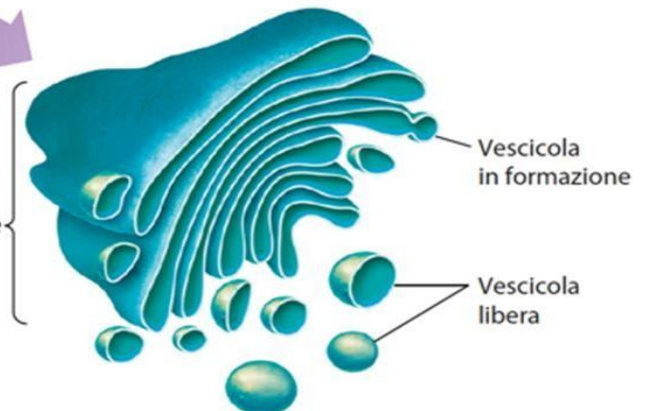
Golgi apparatus



(b) Apparato del Golgi in una cellula della radice di fagiolo

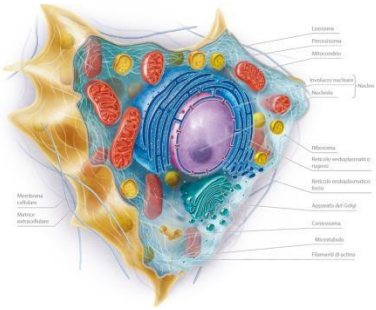
0,2 μm

FIGURA 4.18 Apparato del Golgi. (a) Sezione che mostra la disposizione e le dimensioni relative dell'apparato del Golgi all'interno di una cellula. (b) Fotografia al microscopio elettronico a trasmissione dell'apparato del Golgi di una cellula dell'apice radicale di fagiolo (TEM). (c) Rappresentazione schematica dell'apparato del Golgi che evidenzia la formazione di vescicole per gemmazione.



(c) Apparato del Golgi

The cell: basic unit of life

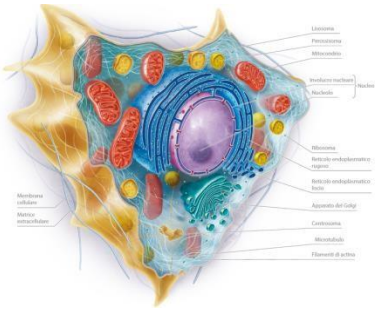


Golgi apparatus

The Golgi complex is like a processing station

- The contents of vesicles from the ER are modified and processed in the Golgi complex
- E.g., secretory and membrane proteins are mainly **glycosylated** (the addition of short-chain carbohydrates), a process that begins in the ER and is completed in the Golgi complex
- The processed substances then move to other locations in the cell through vesicles that bud off of the Golgi complex

The cell: basic unit of life



Golgi apparatus

- Once processed by the Golgi complex, materials to be exported from the cell are packaged into **secretory vesicles**
- These move to the plasma membrane and fuse with it, releasing their contents outside the cell
- The ER, Golgi, secretory vesicles and lysosomes make up the **endomembrane system** of the cell, responsible for **trafficking** substances through the cell

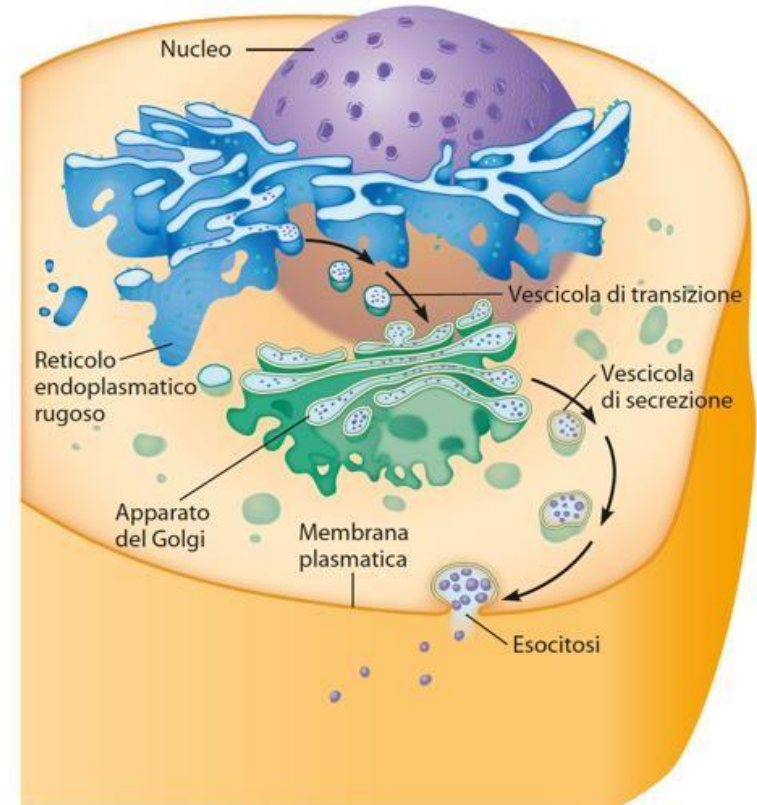
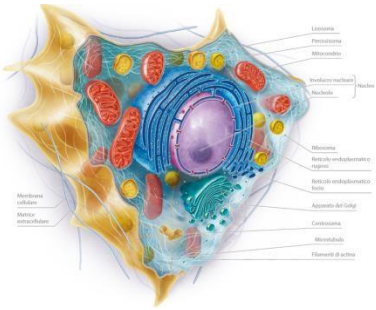


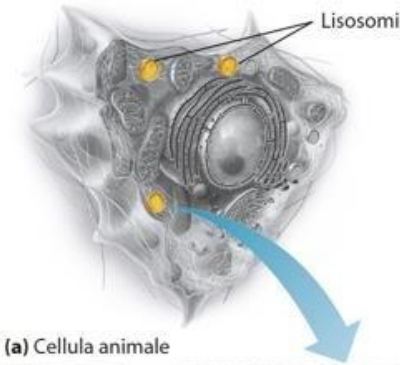
FIGURA 4.19 Processo di secrezione nelle cellule eucariote. Le proteine destinate all'esportazione vengono sintetizzate sul RE rugoso, trasferite all'apparato del Golgi per la maturazione e infine impacchettate nelle vescicole di secrezione. Queste vescicole si avviano poi verso la membrana plasmatica e, fondendosi con essa, rilasciano il loro contenuto all'esterno della cellula per esocitosi.

The cell: basic unit of life

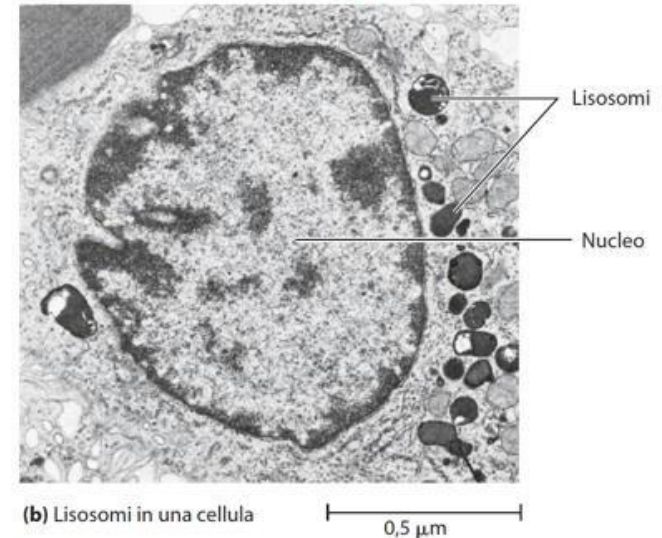


Lysosomes

- **Lysosomes** are single membrane organelles that store *hydrolases*, enzymes that can digest any kind of biological molecule
- These enzymes are sequestered to prevent them from digesting the contents of the cell
- A special carbohydrate coating on the inner lysosome membrane protects it from digestion



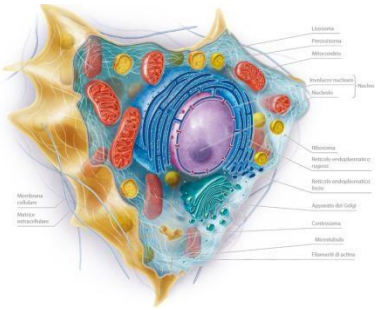
(a) Cellula animale



(b) Lisosomi in una cellula

FIGURA 4.20 Lisosomi. (a) Una sezione che mostra i lisosomi all'interno di una cellula animale. (b) Lisosomi in una cellula animale colorati con una tecnica citochimica per la fosfatasi acida, un enzima lisosomiale. La tecnica di colorazione citochimica produce un denso deposito di fosfato di piombo nei siti in cui esiste l'attività della fosfatasi acida (TEM).

The cell: basic unit of life



Peroxisomes

- **Peroxisomes** resemble lysosomes in size and appearance
- They are surrounded by a single Membrane
- Peroxisomes are especially prominent in the liver and kidney cells of animals
- H_2O_2 is highly toxic to cells but can be transformed into water and oxygen by the enzyme *catalase*
- Eukaryotic cells have metabolic processes that produce H_2O_2
- These reactions are confined to peroxisomes that contain catalase, so that cells are protected from the harmful effects of peroxide

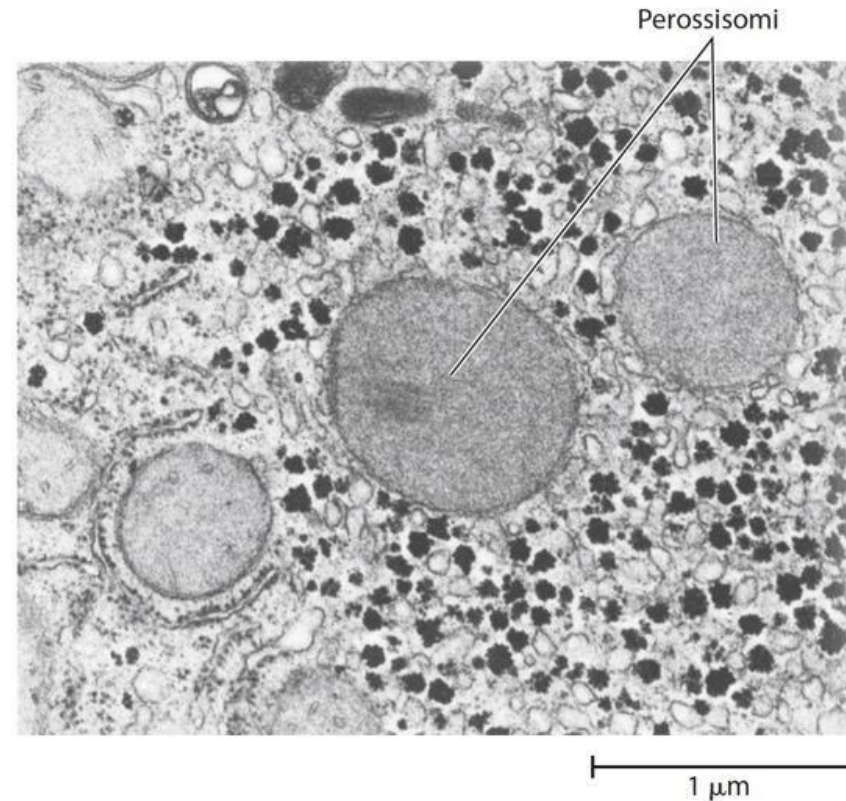
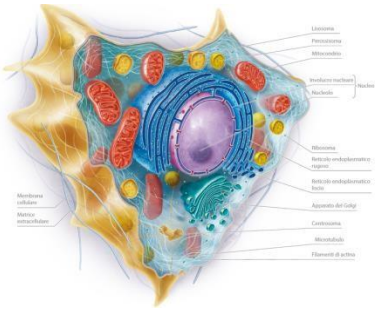
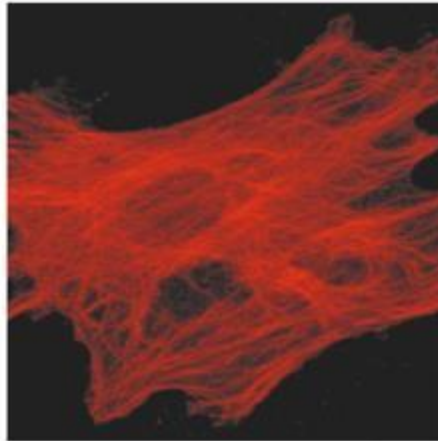


FIGURA 4.21 Perossisomi animali. In questa sezione trasversale di una cellula di fegato si possono notare diversi perossisomi (TEM). I perossisomi si trovano nella maggior parte delle cellule animali, ma sono particolarmente abbondanti nelle cellule epatiche e renali.

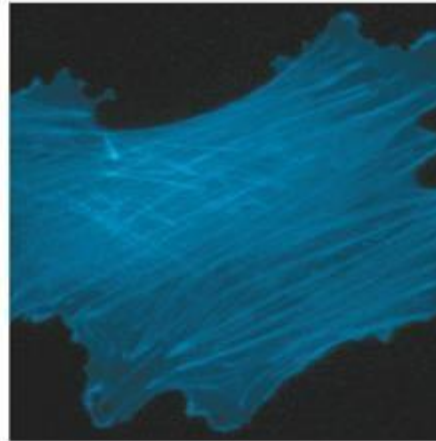
The cell: basic unit of life



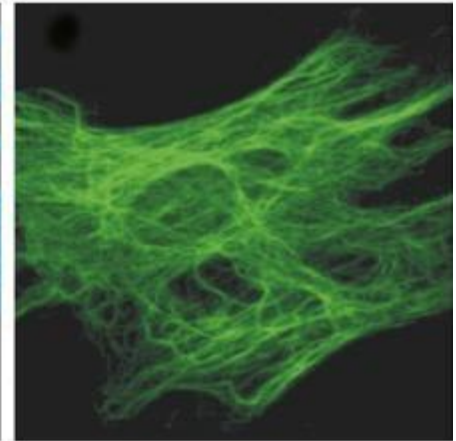
Cytoskeleton



(a) Microtubuli



(b) Microfilamenti



(c) Filamenti intermedi

10 μm

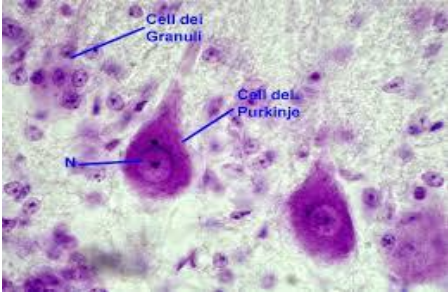
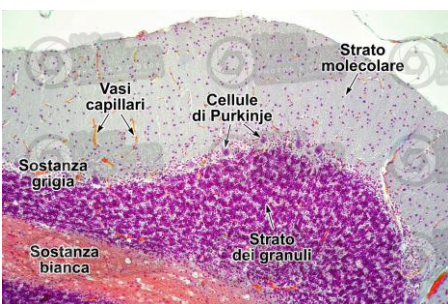
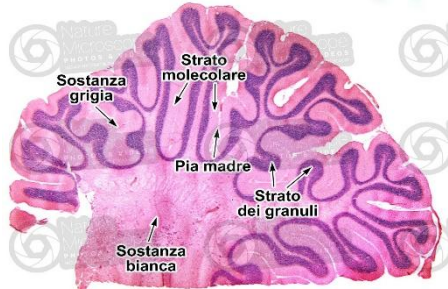
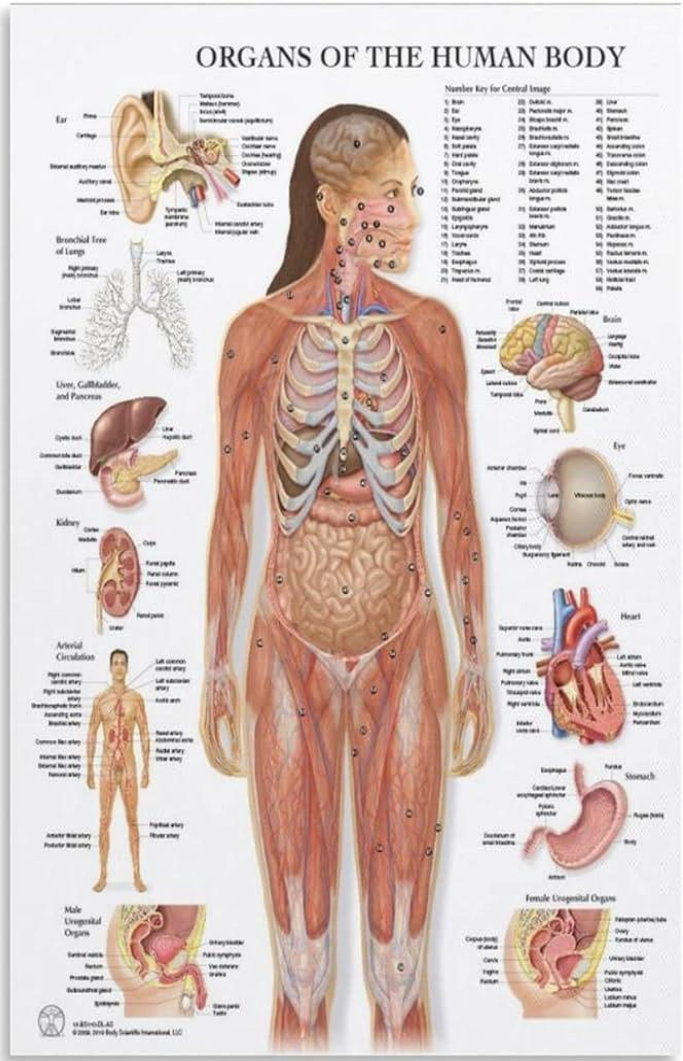
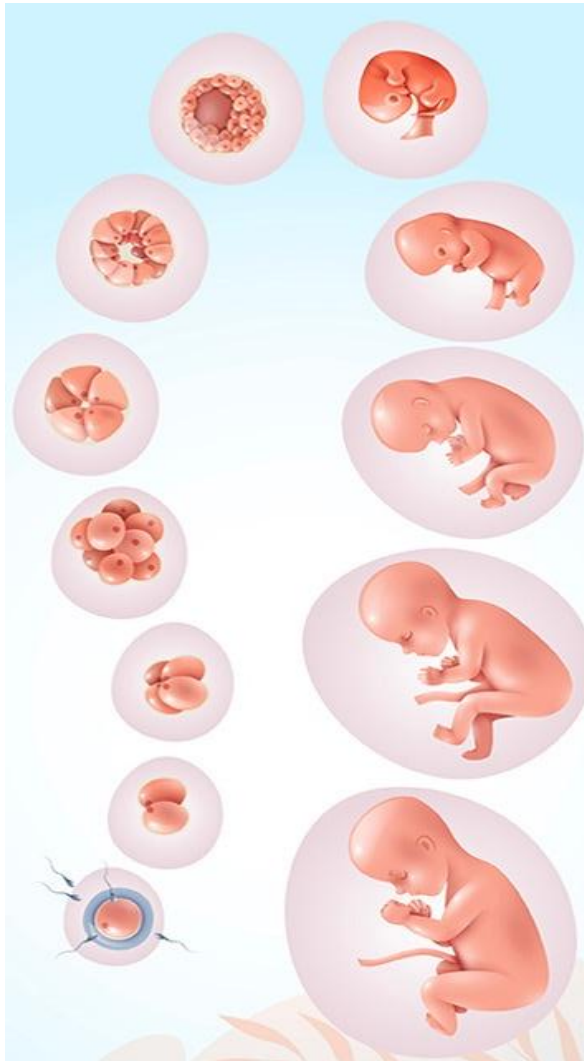
- The cytoskeleton *is a three-dimensional array of interconnected microfilaments, microtubules and intermediate filaments*
- It gives a cell its *distinctive shape and internal organization*
- It also plays a role in cell movement and cell division

FIGURA 4.25

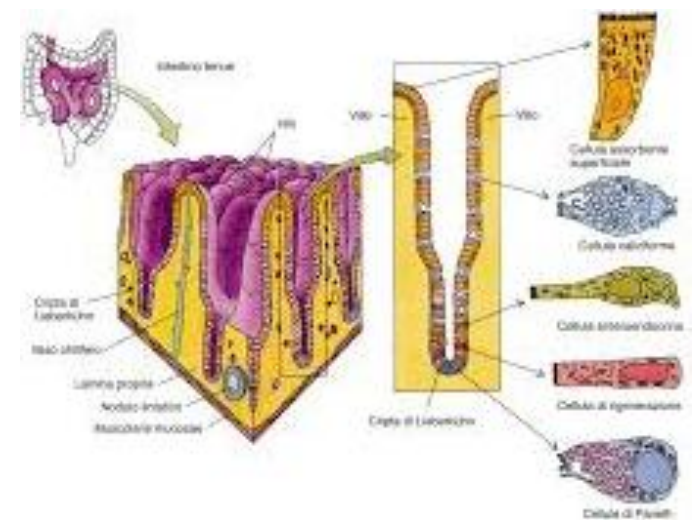
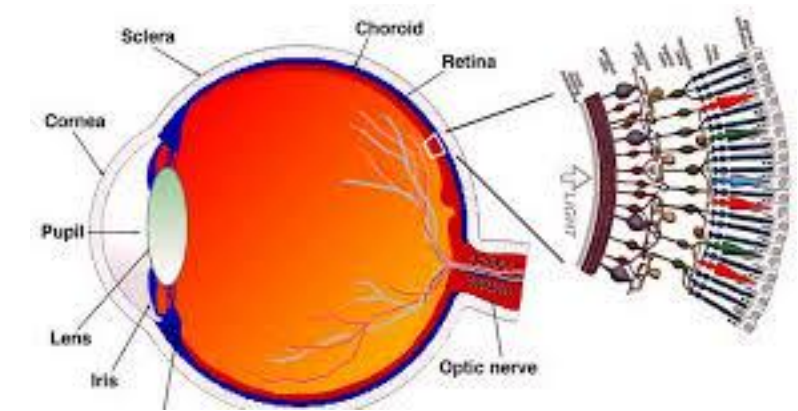
Citoscheletro. Questa micrografia utilizza la tecnica della microscopia a fluorescenza per evidenziare il citoscheletro di osteoblasti (cellule che producono le ossa) (Capitolo 15). In queste tre immagini della stessa cellula, un anticorpo fluorescente riconosce (a) i microtubuli (rosso), un altro (b) i microfilamenti (blu) e un terzo (c) i filamenti intermedi (verde).

The cell: basic unit of life

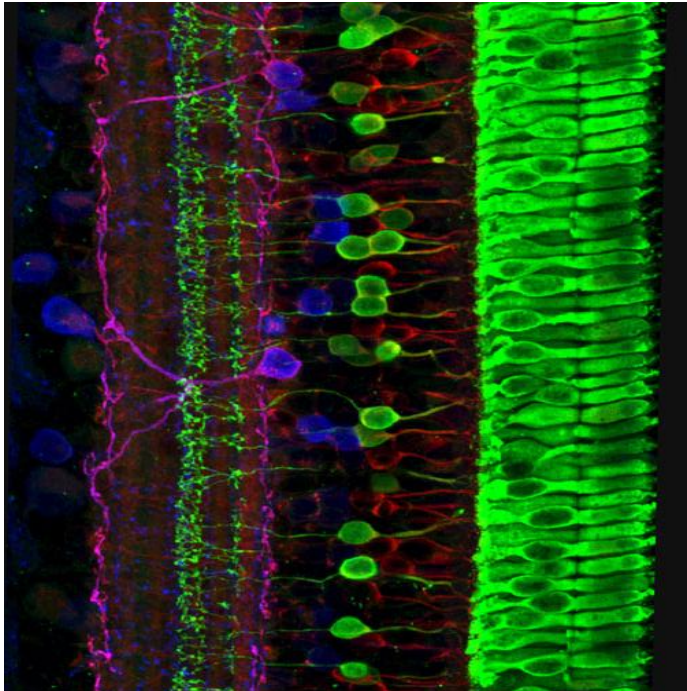
Si stima che un essere umano adulto abbia tra 30 e 40 trilioni di cellule



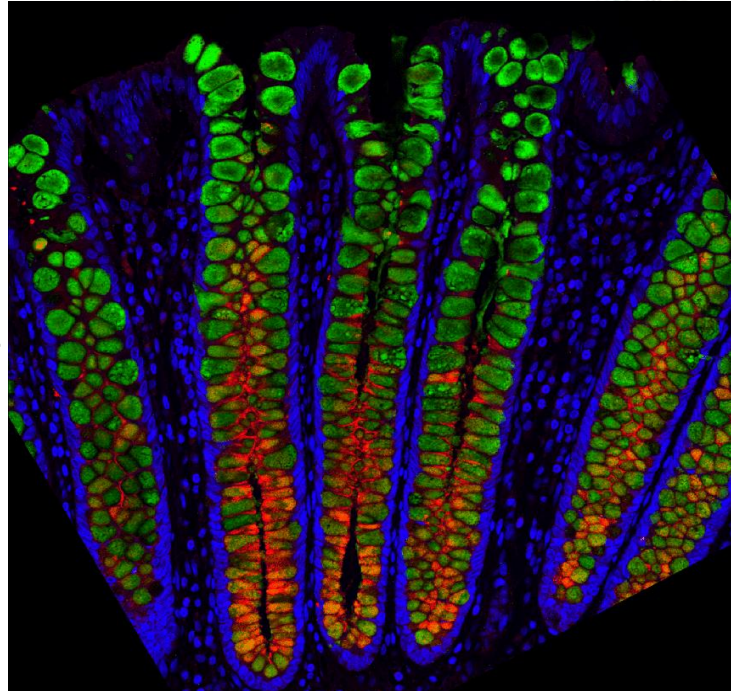
The cell: basic unit of life



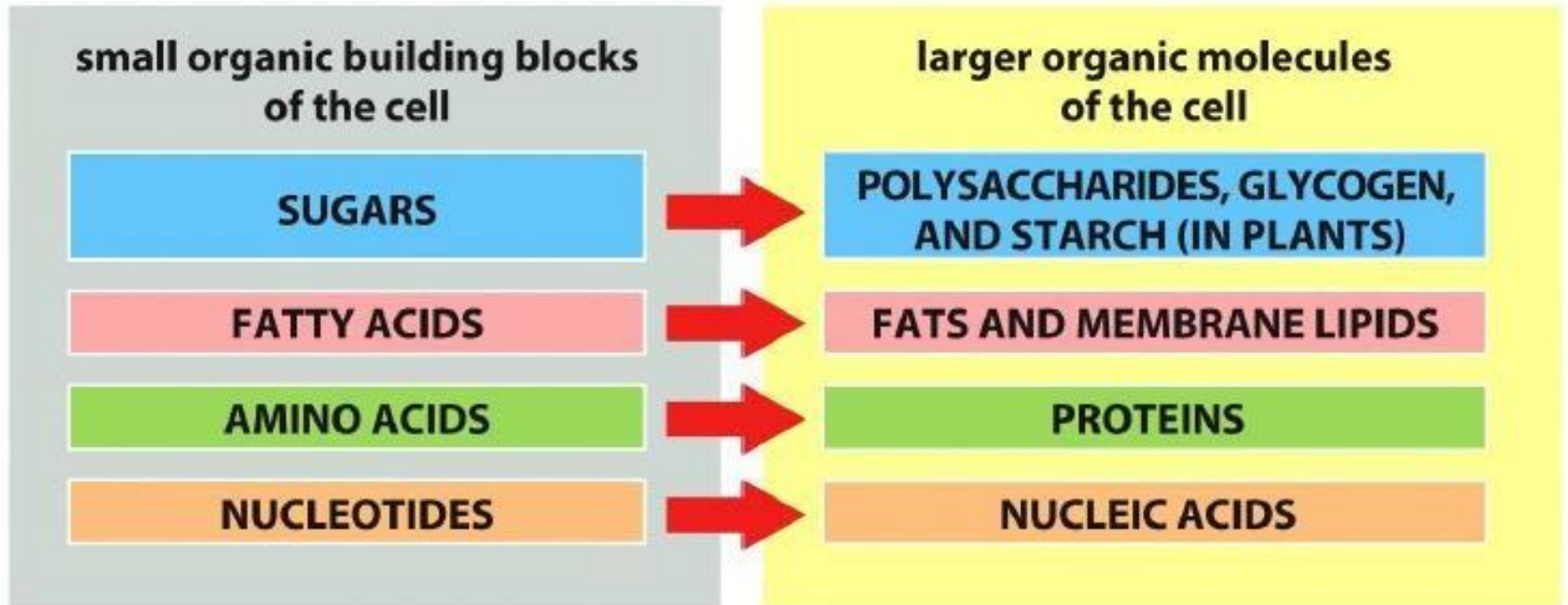
Cellule della Retina



Cellule delle Cripte Intestinali

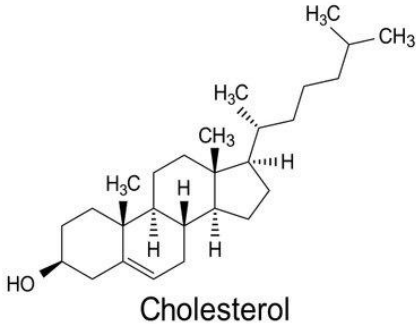
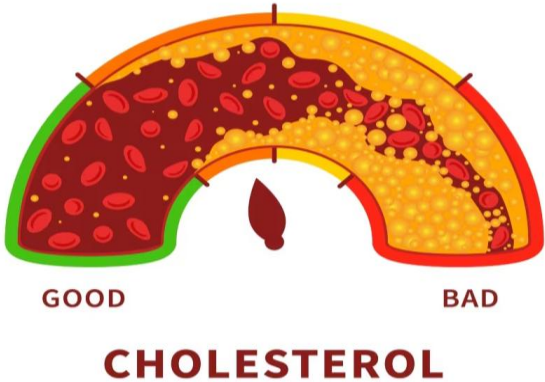


The macromolecules of the cell

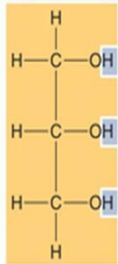


Lipidi - Grassi

- Composti principalmente da atomi di idrogeno (H) e carbonio (C)
- I lipidi sono non-polari e quindi insolubili in acqua

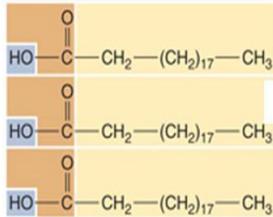


Gli atomi di idrogeno di ogni gruppo ossidrilico del glicerolo vengono rimossi



Glicerolo

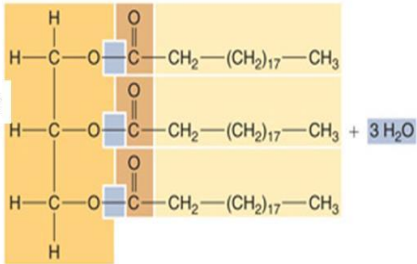
I gruppi ossidrilici di ogni gruppo carbossilico dei 3 acidi grassi vengono rimossi



3 acidi grassi

I nuovi legami formati sono chiamati legame di estere

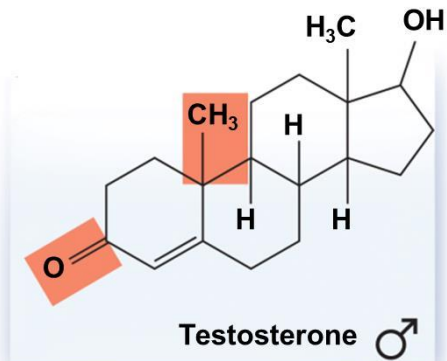
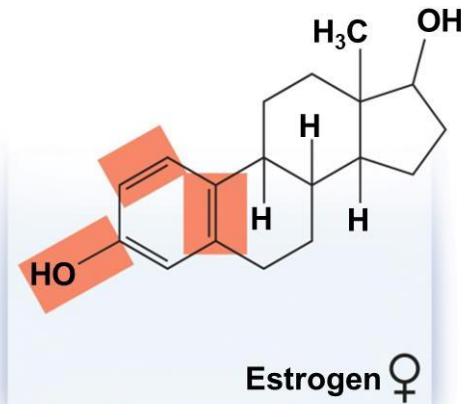
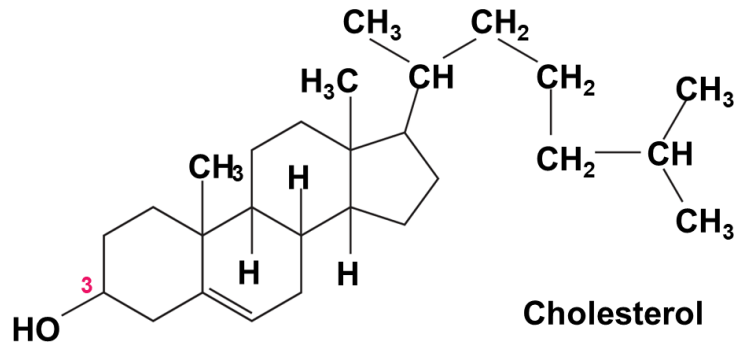
disidratazione



Trigliceride (grasso)

Lipidi - Grassi

Ormoni steroidei



Il **colesterolo** è la molecola da cui originano gli ormoni sessuali:
Estrogeni e Androgeni



Female cardinal



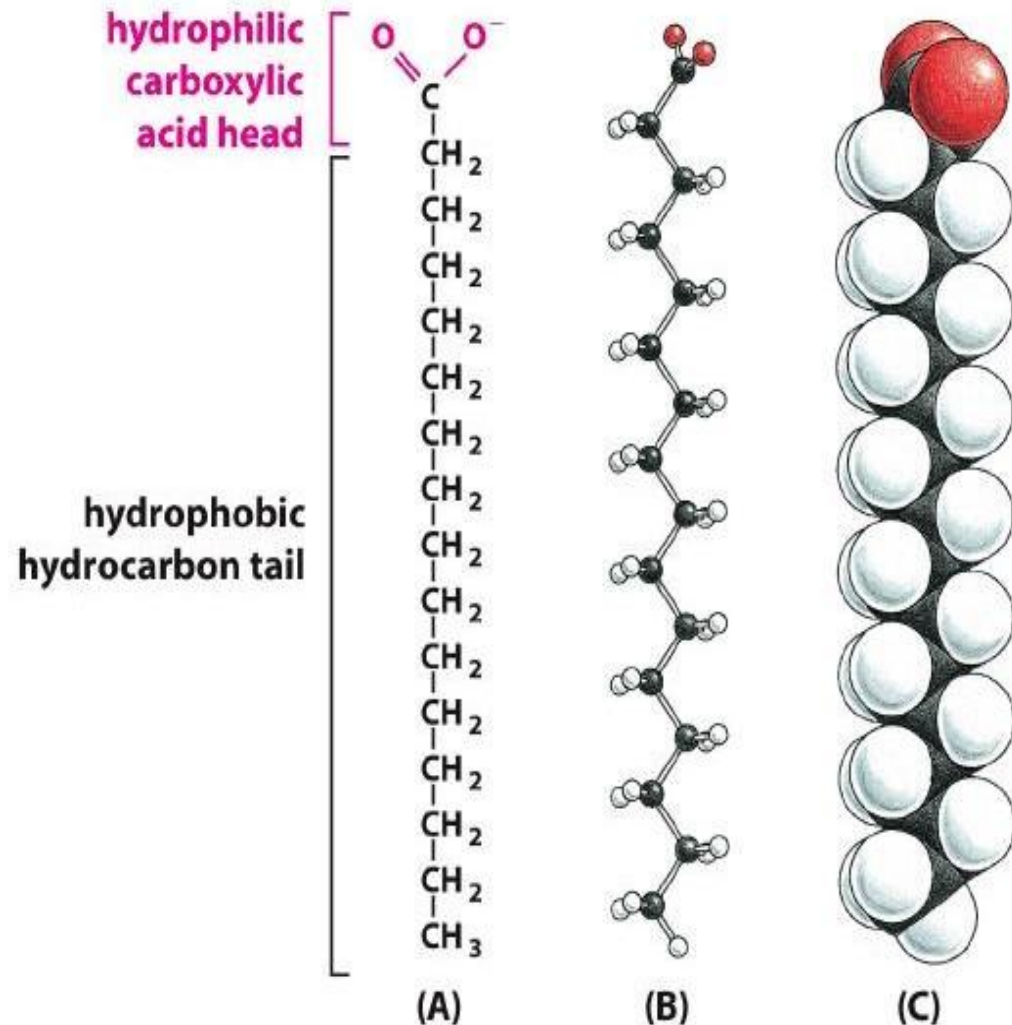
Male cardinal

Lipidi - Grassi

A parte infarto ed ictus.....

I lipidi sono fondamentali per:

1. La generazione della cellula stessa e dei vari organelli grazie al loro ruolo cruciale nella formazione delle membrane.
2. Immagazzinare energia, 1 grammo di grasso immagazzina due volte più energia di 1 grammo di glicogeno o amido.



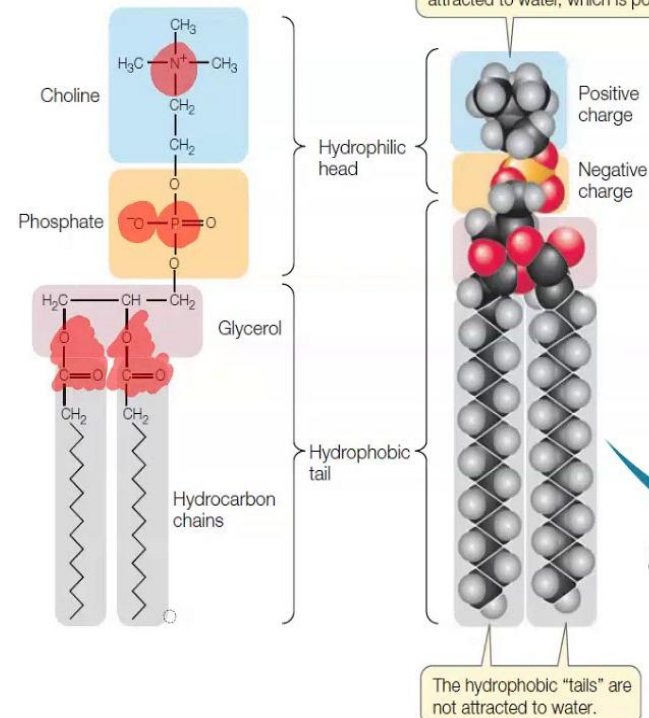
Lipidi - Grassi

FOSFOLIPIDI

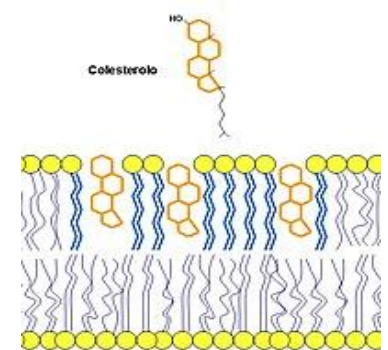
Struttura: 2 acidi grassi, glicerolo ed un gruppo fosfato al quale si legano specifiche molecole quali colina, inositolo, serina, etanolammina.

1. Regione del fosfato è polare ed idrofilica
2. Catene di acidi grassi sono apolari ed idrofobici

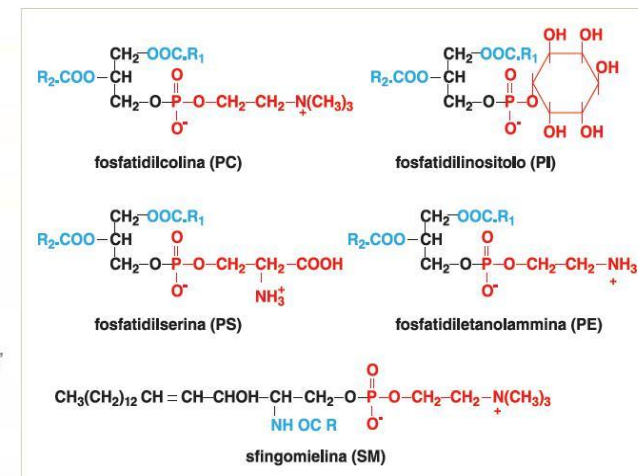
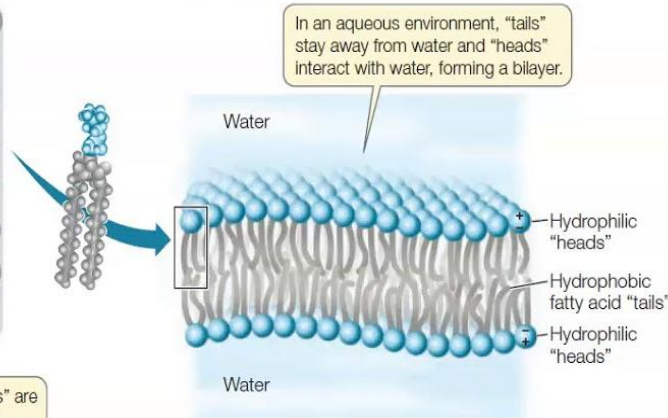
(A) Phosphatidylcholine



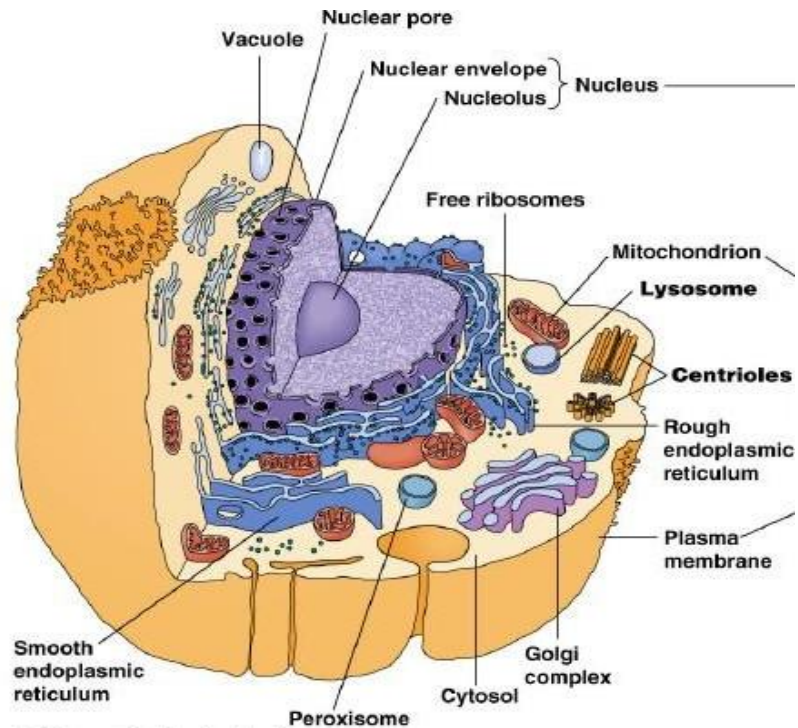
Il colesterolo aumenta l'ordine del tratto prossimale delle catene idrofobiche, diminuendo l'isomerizzazione trans-gauche e gli inginocchiamenti transitori delle catene stesse.



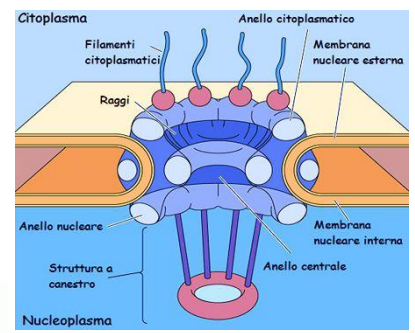
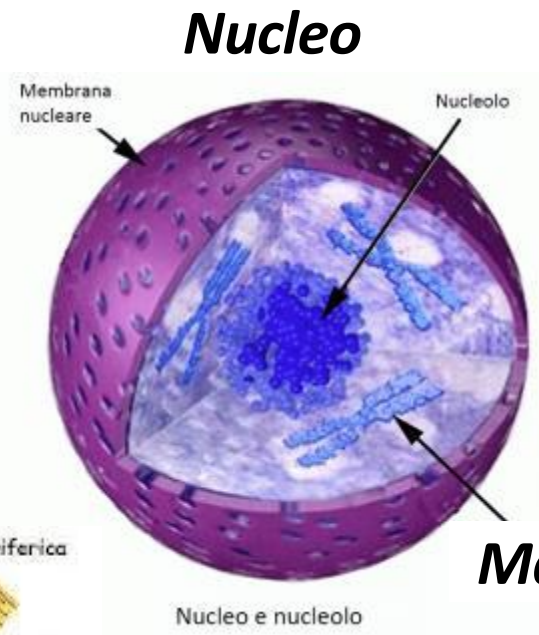
(B) Phospholipid bilayer



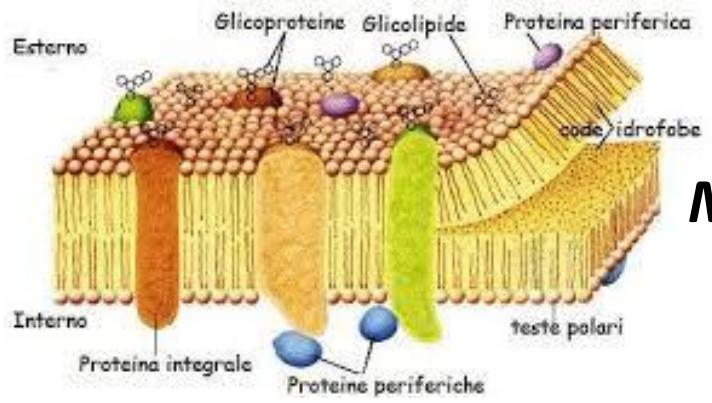
Eucarioti: 'vero' nucleo = circondato da membrana



(a) Generalized animal cell
© 2012 Pearson Education, Inc.



Poro Nucleare

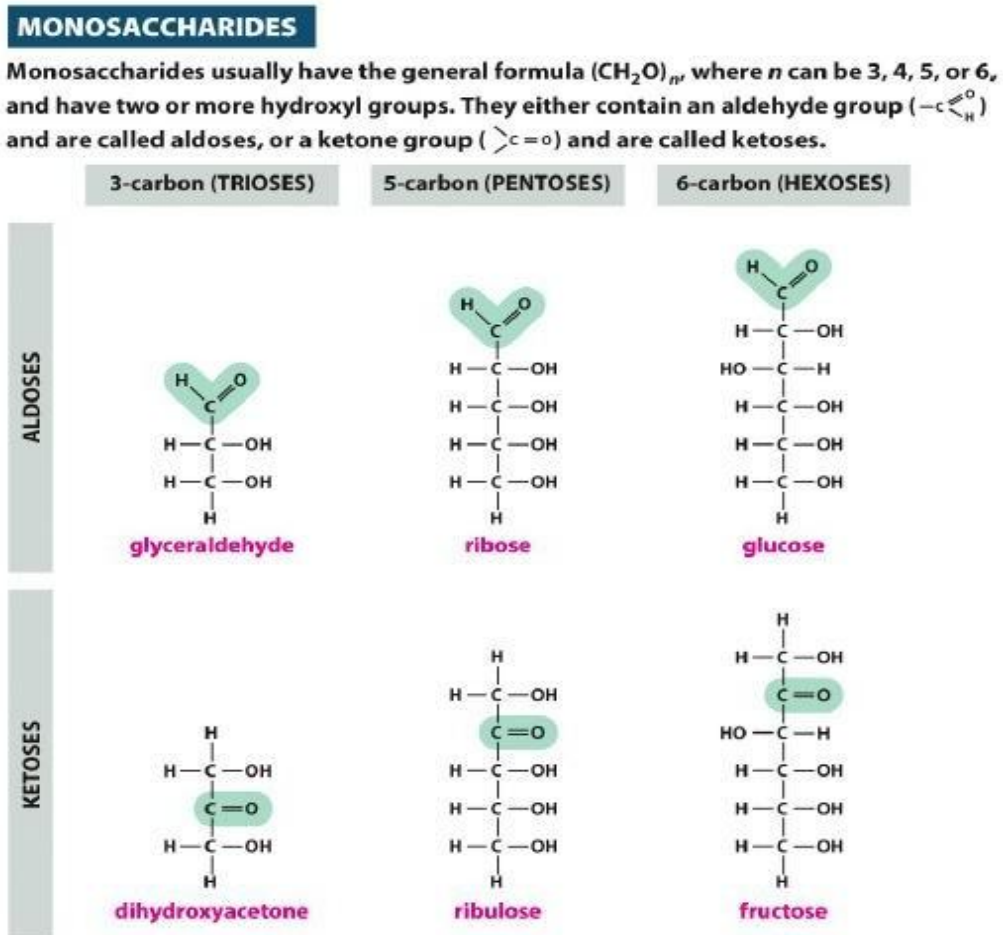


Membrana Nucleare

Carboidrati

I carboidrati, detti anche glucidi (dal greco "glucos" = dolce) sono sostanze formate da carbonio ed acqua e sono contenuti principalmente negli alimenti di origine vegetale.

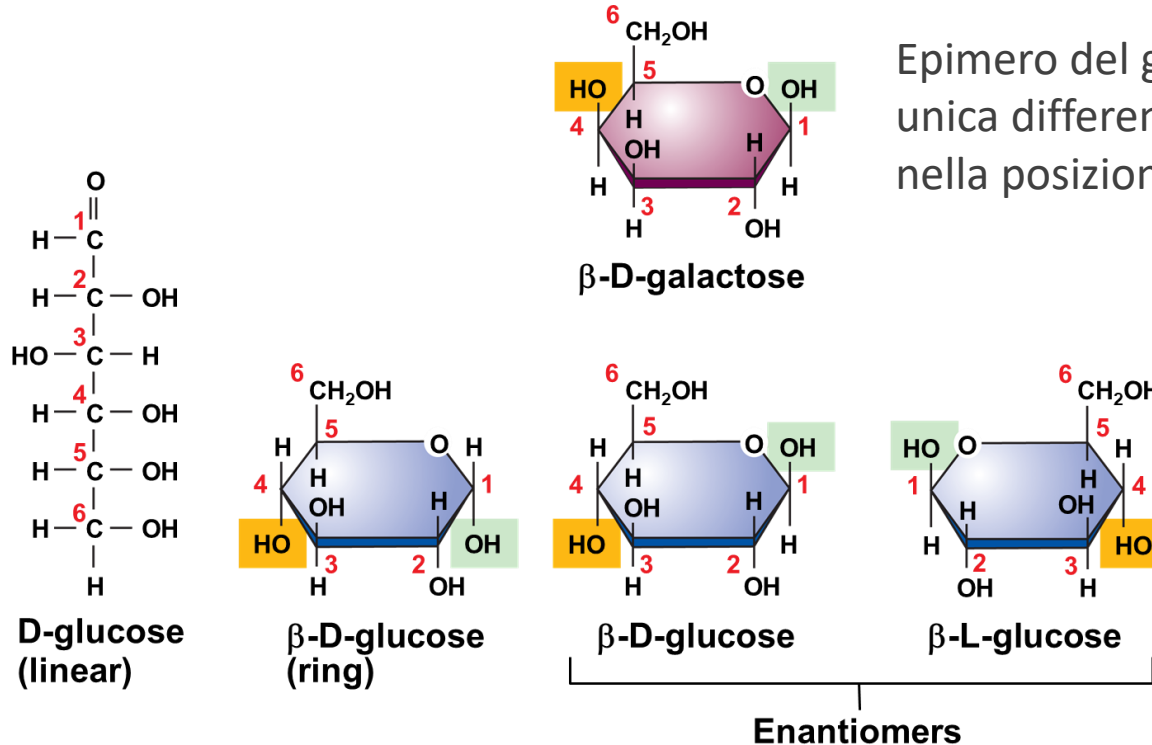
La forma più semplice di carboidrati sono i monosaccaridi o zuccheri semplici come glucosio, fruttosio, ribosio $C_n(H_2O)_n$. Sono molecole polari e quindi facilmente solubili in acqua. In base alla loro struttura chimica si distinguono in Aldosi e Chetosi.



Carboidrati

RING FORMATION

In aqueous solution, the aldehyde or ketone group of a sugar molecule tends to react with a hydroxyl group of the same molecule, thereby closing the molecule into a ring.



Epimero del glucosio:
unica differenza fra i due esosi è
nella posizione dell'-OH sul C-4.

(a) Linear and ring structures
of α -D-glucose

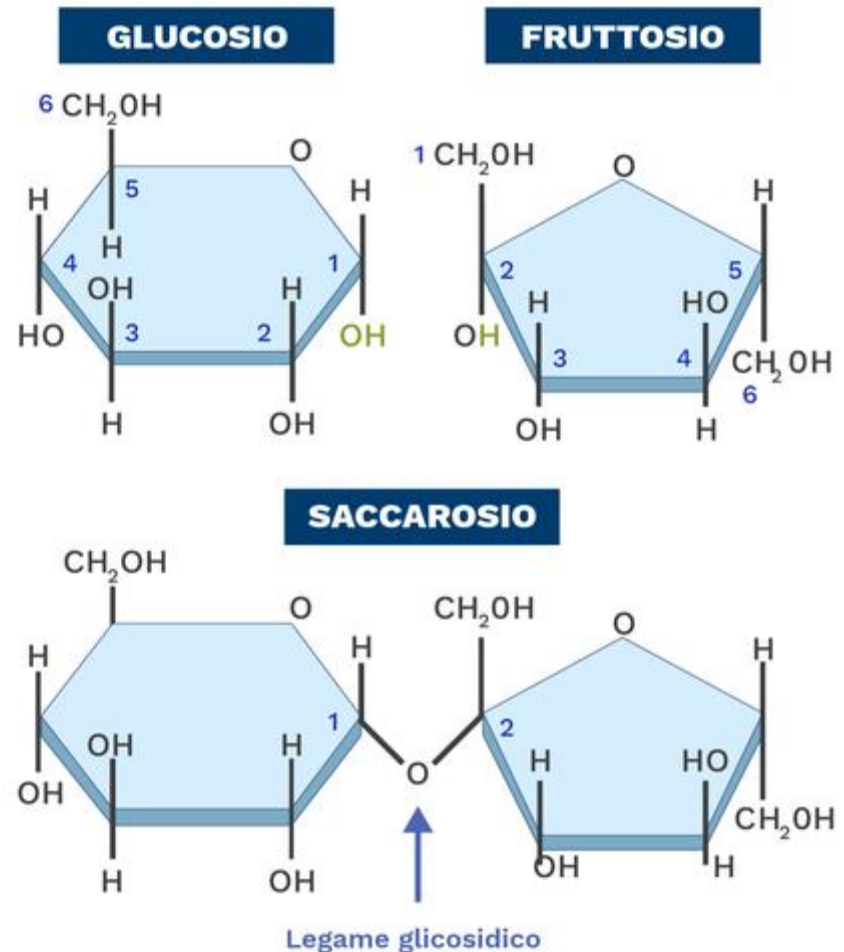
(b) Isomers of glucose

Carboidrati

Disaccaridi

Per formare un disaccaride due monosaccaridi si uniscono tramite un legame glicosidico tra il carbonio di un monosaccaride e il gruppo alcolico dell'altro monosaccaride con eliminazione di una molecola di acqua.

I **più comuni disaccaridi** sono: il **saccarosio**, zucchero da tavola (sintetizzato dalle piante attraverso il processo di fotosintesi clorofilliana) è costituito da una molecola di glucosio + una molecola di fruttosio; il **lattosio** (si trova nel latte di ogni mammifero) è costituito da una molecola di glucosio + una molecola di galattosio; il **maltosio**, costituito da due molecole di glucosio.

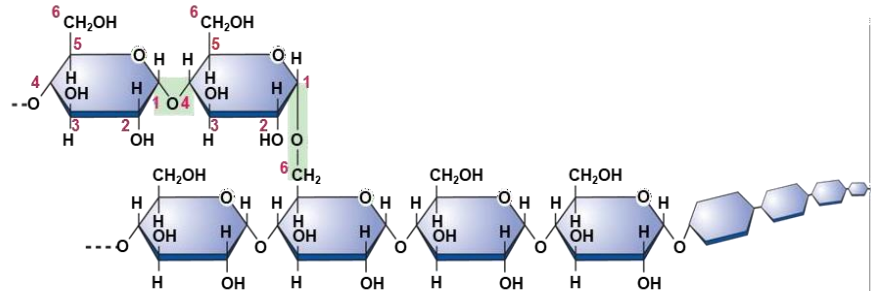


Carboidrati

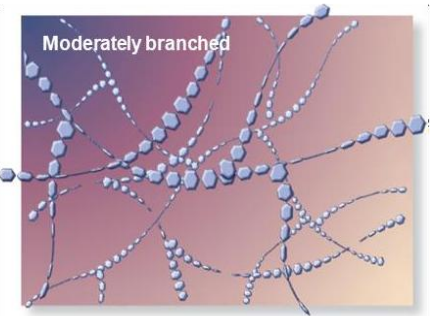
Polisaccaridi

Poco ramificato
poco solubile in H_2O

Starch [Amido]

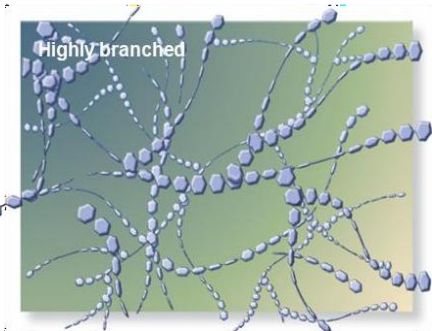
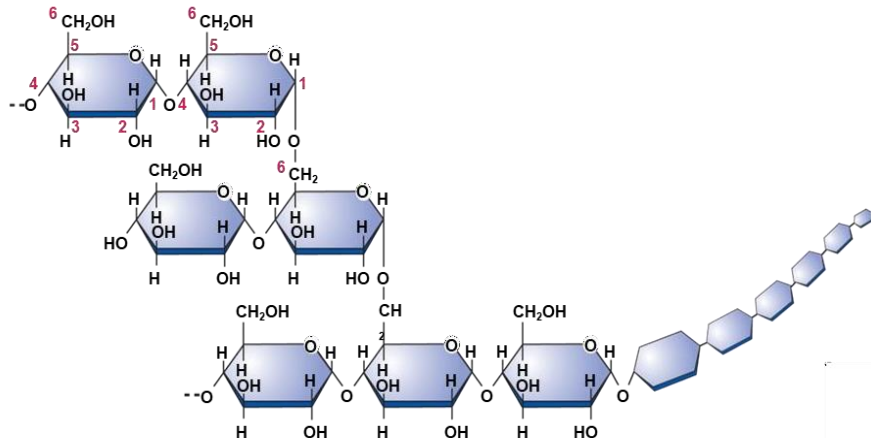


Branching patterns



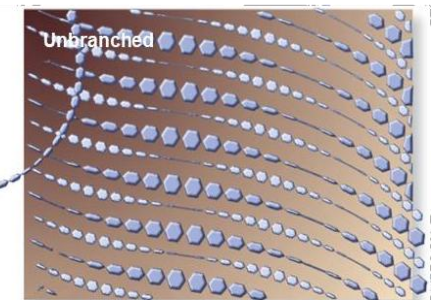
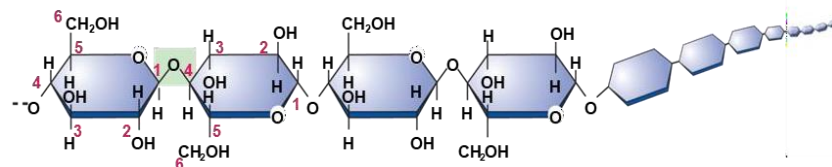
Molto ramificato
solubile in H_2O

Glycogen [Glicogeno]



Non ramificato
insolubile in H_2O

Cellulose [Cellulosa]



Acidi nucleici

Responsabile dell'immagazzinamento, trasmissione ed utilizzo dell'informazione genetica.

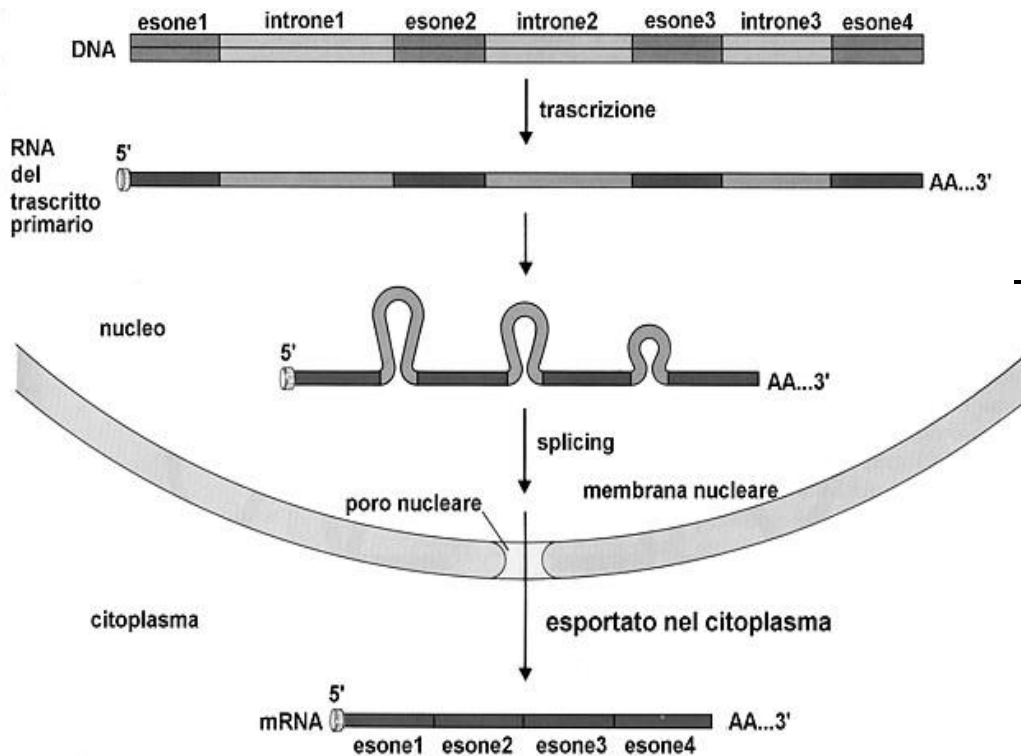
Se ne distinguono due classi:

- Acido deossiribonucleico (DNA)

Relegato all'interno del nucleo, immagazzina le informazioni genetiche codificate nella sequenza dei suoi monomeri

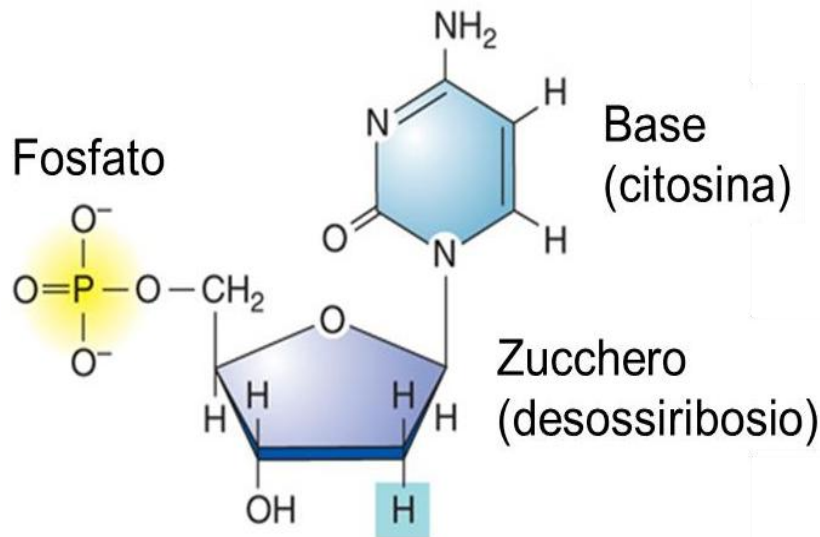
- Acido ribonucleico (RNA)

Coinvolto nella trasmissione di queste informazioni all'esterno del nucleo rendendole istruzioni funzionali per la cellula

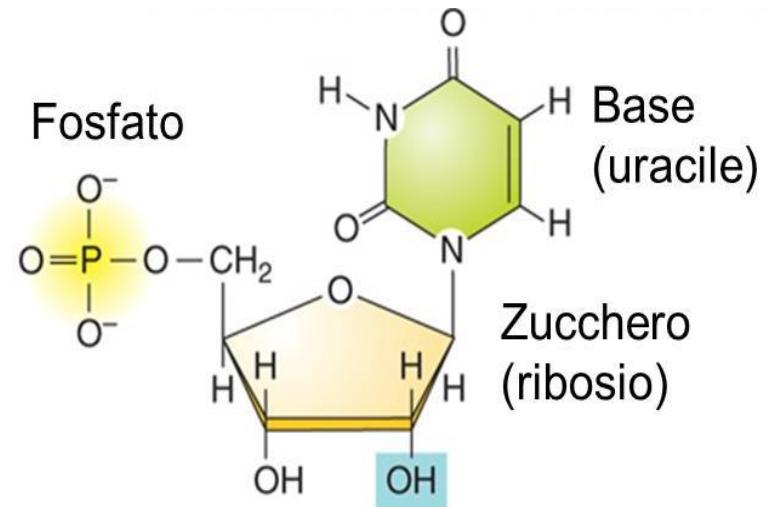


Acidi nucleici- I Nucleotidi

Il monomero degli acidi nucleici è il nucleotide che è costituito da:
uno zucchero a cinque atomi di carbonio (desossiribosio DNA e ribosio RNA), una base formata da un singolo o doppio anello contenente azoto (base azzotata), un gruppo fosfato.



Esempio di un desossiribonucleotide

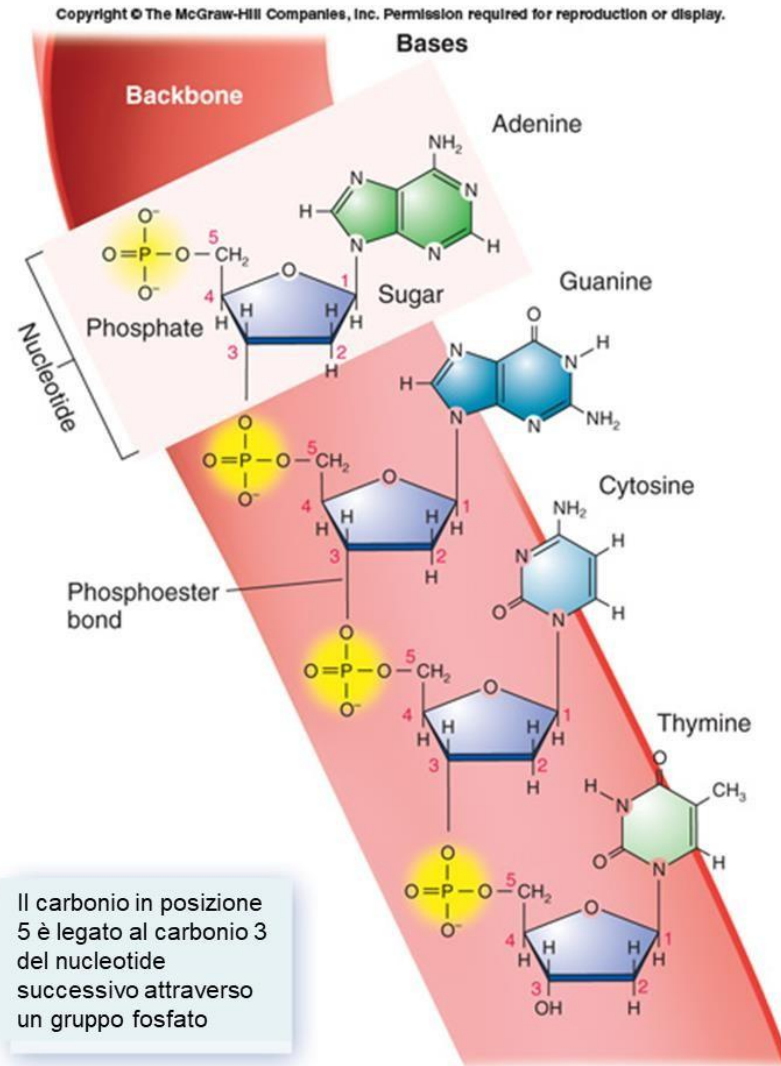
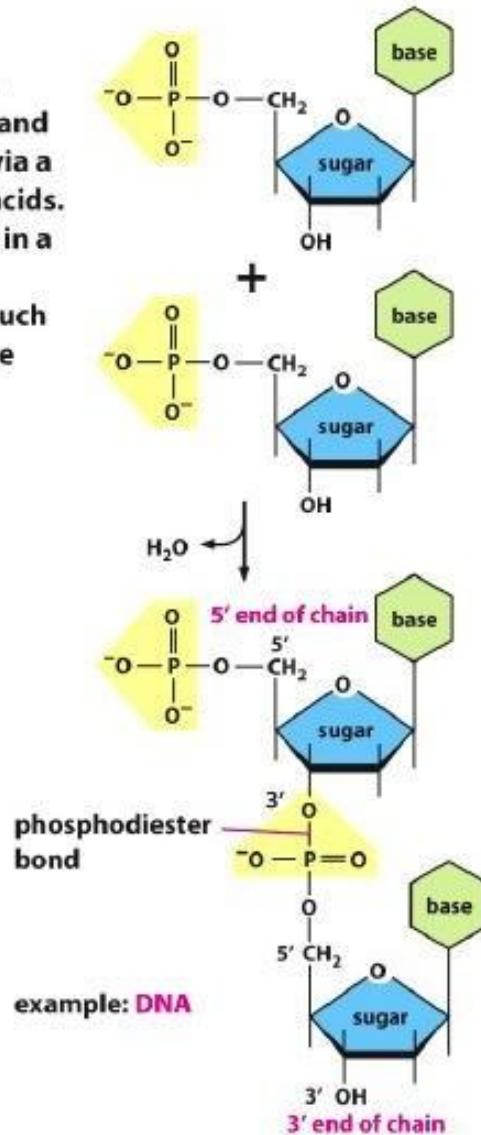


Esempio di un ribonucleotide

Acidi nucleici- Il Legame Fosfodiester

NUCLEIC ACIDS

Nucleotides are joined together by **phosphodiester bonds** between 5' and 3' carbon atoms of the sugar ring, via a phosphate group, to form nucleic acids. The linear sequence of nucleotides in a nucleic acid chain is commonly abbreviated by a one-letter code, such as AGCTTACA, with the 5' end of the chain at the left.



Il carbonio in posizione 5 è legato al carbonio 3 del nucleotide successivo attraverso un gruppo fosfato

Acidi nucleici- Le Basi Azzotate

NOMENCLATURE

The names can be confusing, but the abbreviations are clear.

BASE	NUCLEOSIDE	ABBR.
adenine	adenosine	A
guanine	guanosine	G
cytosine	cytidine	C
uracil	uridine	U
thymine	thymidine	T

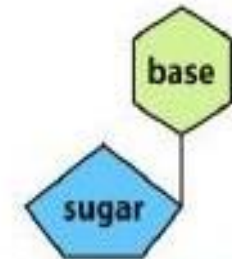
Nucleotides are abbreviated by three capital letters. Some examples follow:

AMP = adenosine monophosphate

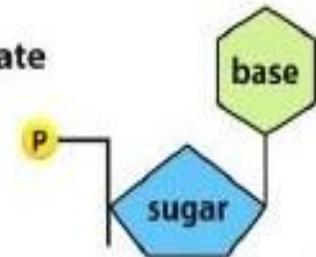
dAMP = deoxyadenosine monophosphate

UDP = uridine diphosphate

ATP = adenosine triphosphate

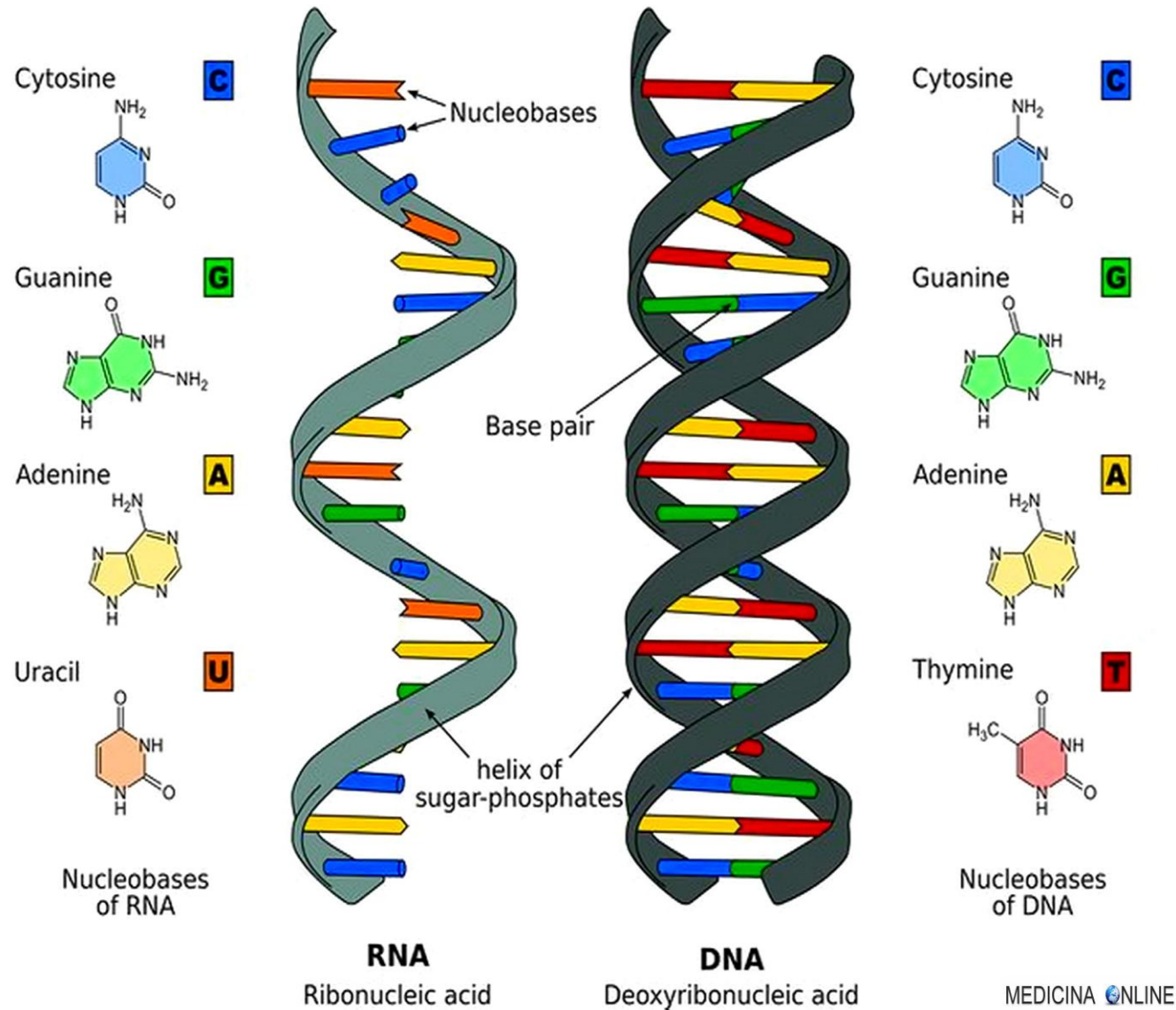


BASE + SUGAR = **NUCLEOSIDE**



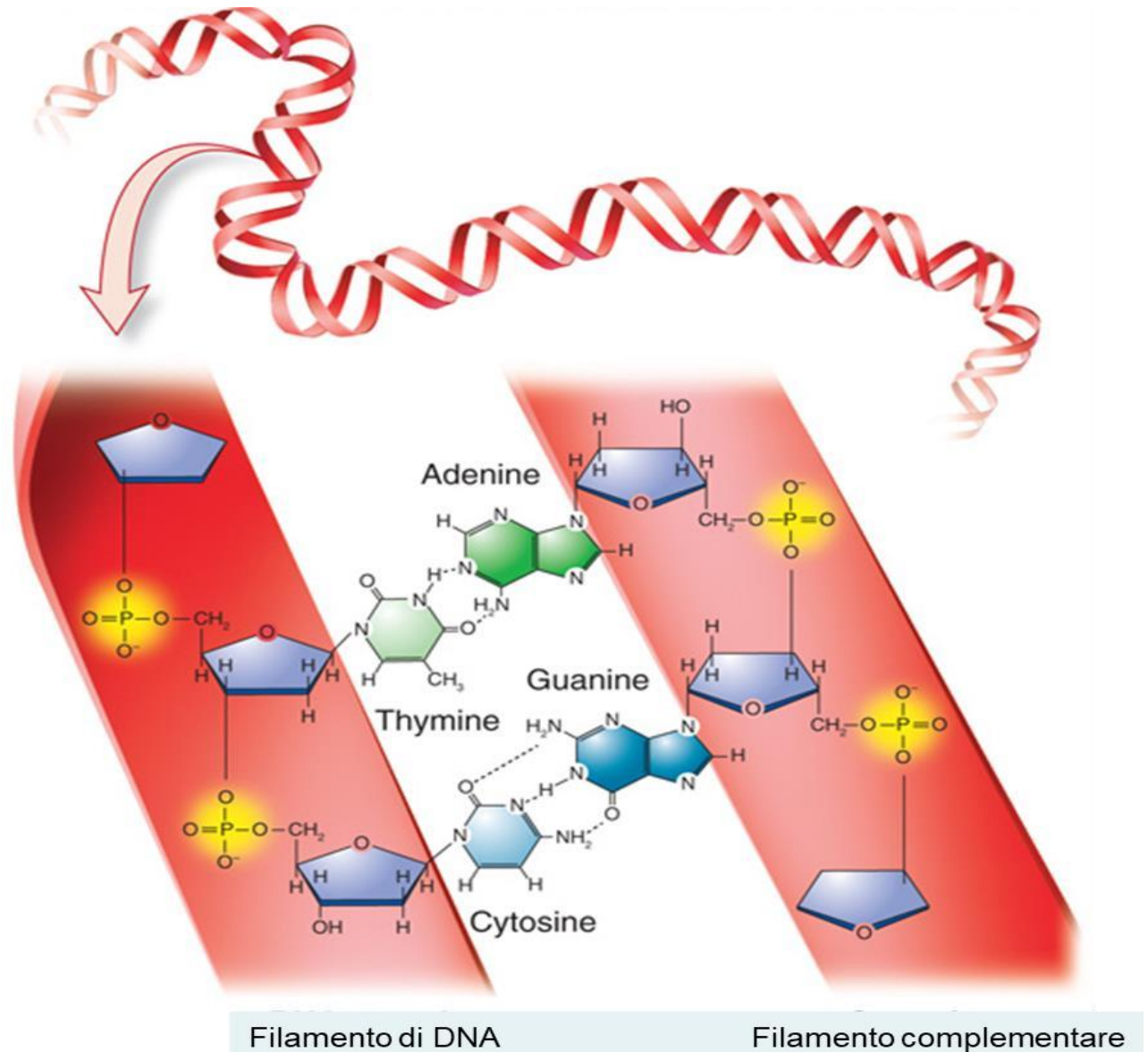
BASE + SUGAR + PHOSPHATE = **NUCLEOTIDE**

Acidi nucleici- Interazioni e Struttura



Acidi nucleici- Interazioni e Struttura

DNA



Acidi nucleici- Interazioni e Struttura

RNA

Strutture secondary e terziarie



mRNA



Encodes proteins

tRNA



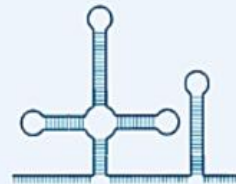
Acts as adaptor between mRNA and amino acids

rRNA



Forms the ribosome

snRNA



Functions in various nuclear processes (e.g. splicing)

snoRNA



Facilitates chemical modification of RNAs

miRNA



Regulates gene expression

siRNA



Silences gene expression

lncRNA



Regulates gene expression

Molteplici funzioni

Proteine

Gli aminoacidi sono i monomeri costituenti le proteine

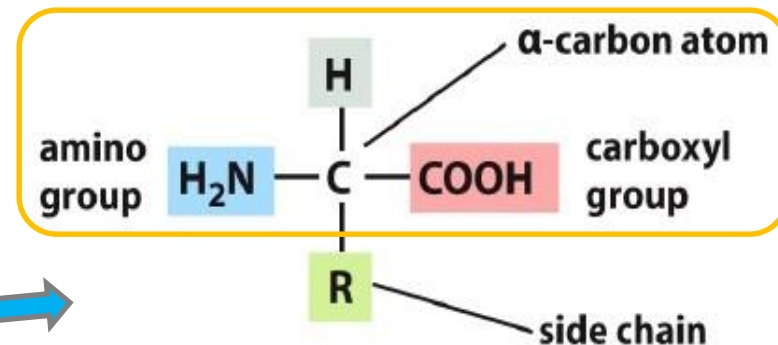
Gli aminoacidi sono 20

Sono composti da:

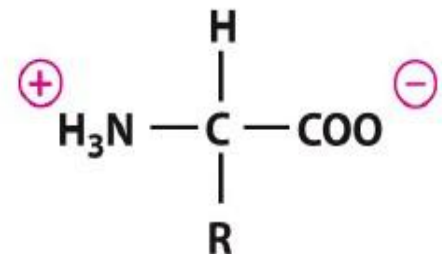
- Una struttura comune
- Gruppi R variabili

THE AMINO ACID

The general formula of an amino acid is



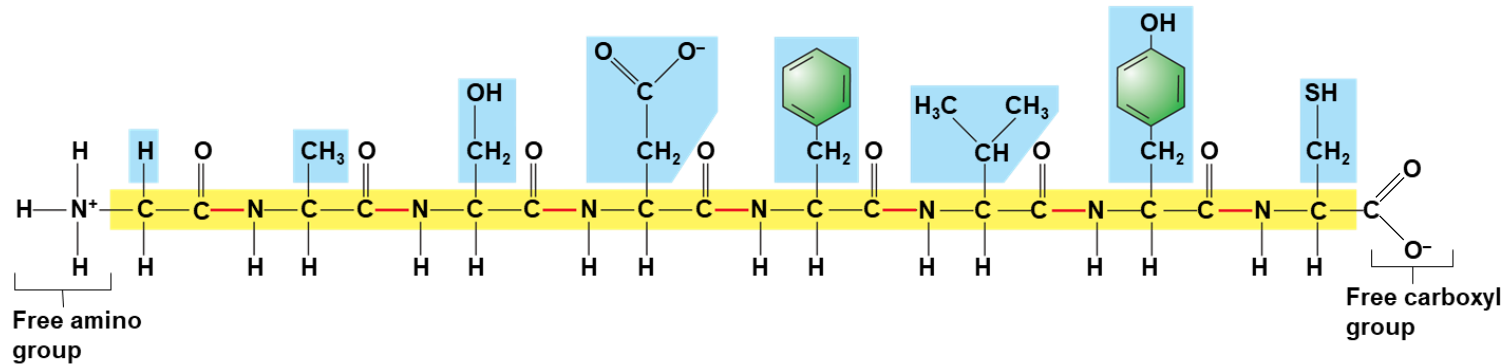
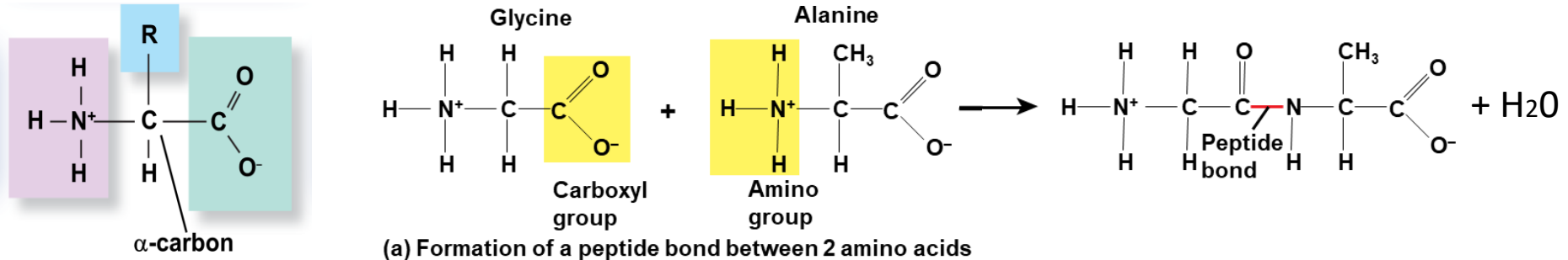
R is commonly one of 20 different side chains. At pH 7, both the amino and carboxyl groups are ionized.



Proteine - Aminoacidi

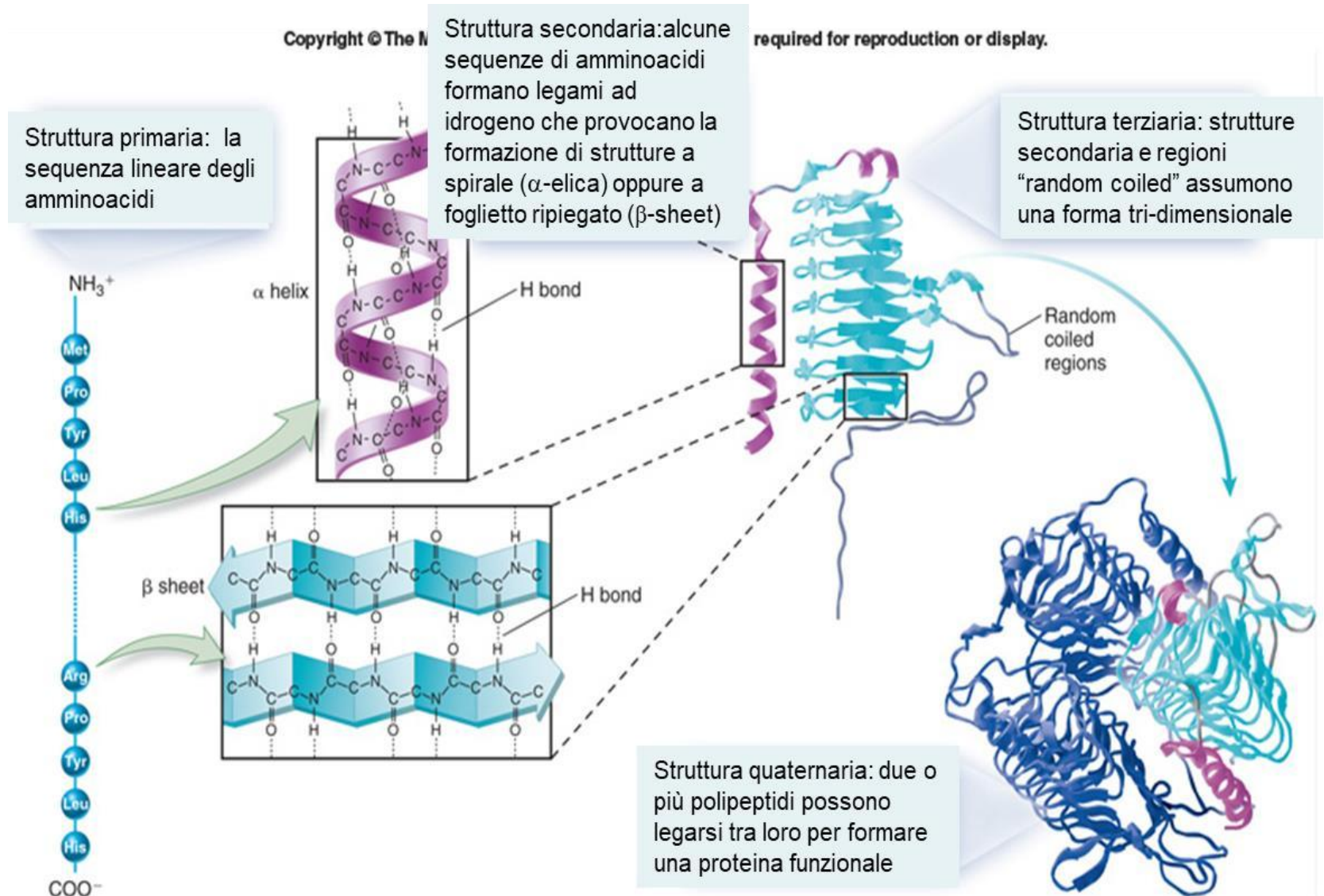


Proteine – Peptidi e Polipeptidi

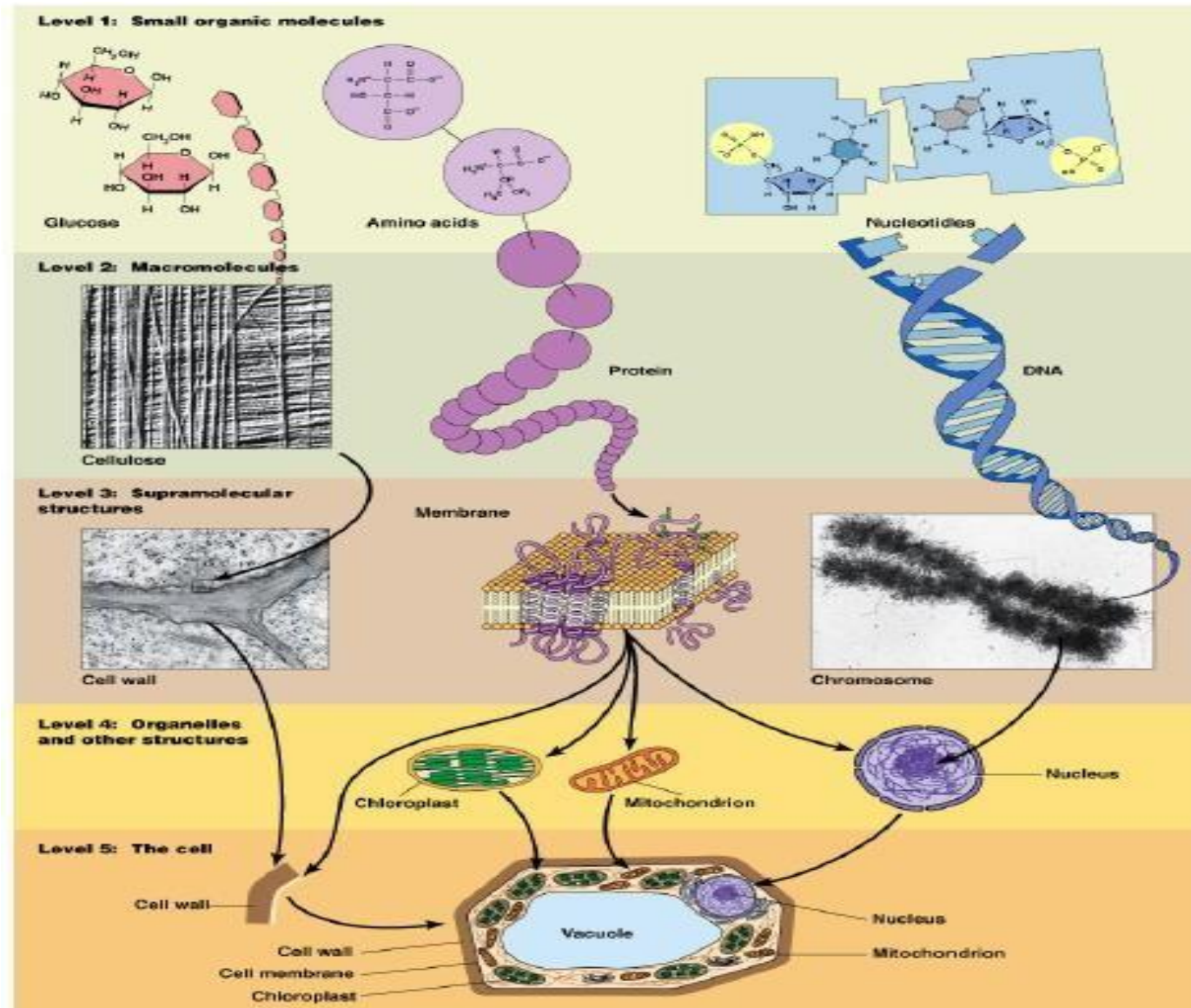


Un **peptide** è costituito da due o più amminoacidi legati tra loro da un legame peptidico a formare una catena composta di norma da 2 – 70 amminoacidi. Catene più lunghe di 70 amminoacidi vengono definite polipeptidi.

Proteine – Strutture dei peptide e polipeptidi

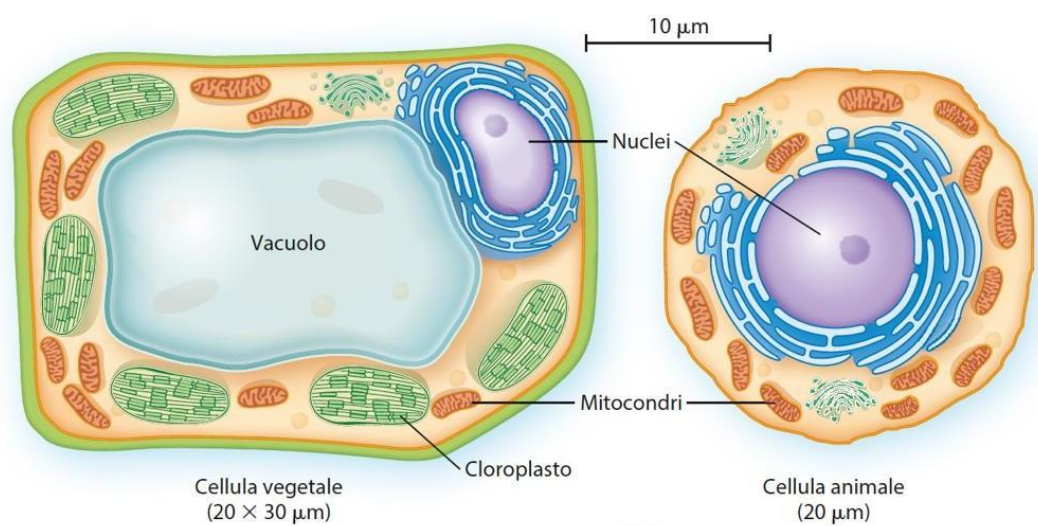


Da molecole semplici a strutture complesse

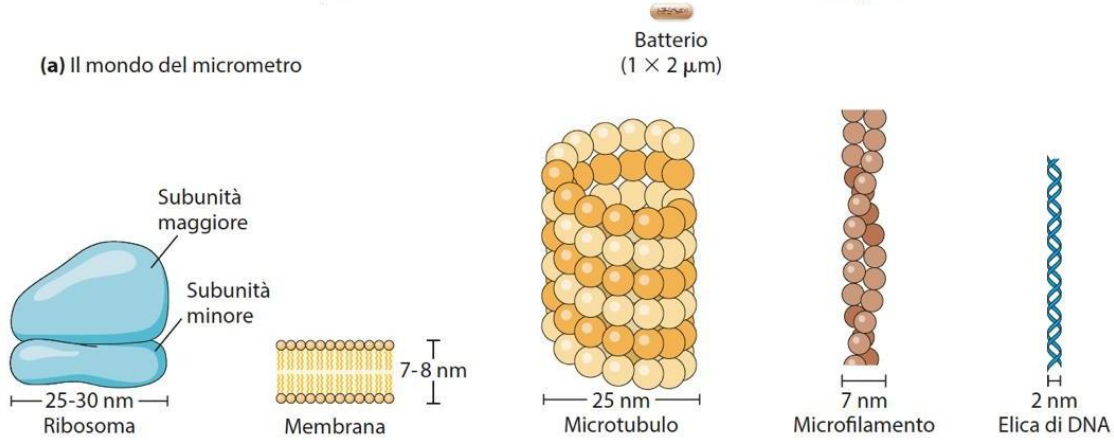


The cell: basic unit of life

Cellule, organelli e molecole: tra micrometri e nanometri



(a) Il mondo del micrometro



(b) Il mondo del nanometro

